Tracing Data Through Analytical General Equilibrium[†]

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Abstract

We develop a heterogeneous-household framework with closed-form aggregate dynamics jointly capturing growth, business cycles, and distributional change. Three moments (aggregate capital and two wealth-sorting covariances) suffice for the equilibrium, which is invariant to household beliefs and transversality. The model is exactly invertible: it takes observed macro and micro data as an equilibrium path and traces the primitives, prices, and distributional states consistent with it. Applied to U.S. aggregates (1950–2019), postwar wealth concentration emerges from return-wealth sorting compounded by TFP growth, and persistent capital-tax reforms produce decades-long distributional drift through the same channel. The business cycle is two-dimensional: TFP is nearly orthogonal to a labor-efficiency and patience block, yielding distinct recession types. PSID and SCF agree on the capital-weighted β within 1.7 pp, and individual β_i is invariant across four postwar tax reforms (OBRA, EGTRRA, ATRA, TCJA).

Keywords: Heterogeneous agents, closed-form aggregation, wealth inequality, business cycles, belief invariance.

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1 Introduction

"...all agree that it began with tracing an outline round a man's shadow."

— Pliny the Elder, *Natural History*, Book XXXV.15 (trans. Rackham, Loeb)

Wealth inequality in the United States has risen steadily since the 1950s, yet the economy has also experienced recurring business cycles and sustained long-run growth. These three phenomena are deeply intertwined. TFP growth raises returns disproportionately for wealthy households with favorable portfolio draws, amplifying inequality through persistent return-wealth sorting. The resulting concentration, in turn, makes the macroeconomy more sensitive to aggregate shocks. Understanding this feedback requires a framework that handles growth, cycles, and distributional dynamics jointly along a non-stationary path. The deeper difficulty is that the forces driving these dynamics (productivity, patience, the labor margin) are never observed directly. We see only the outline they cast in output, capital, hours, and the cross-section of households, and must reconstruct the figure behind it.

This paper constructs such a framework by developing a preference structure that combines familiar ingredients (log utility and a Stone–Geary subsistence level, augmented by a wealth-dependent time endowment) under which the household problem admits closed-form policies that aggregate exactly. Although households face idiosyncratic productivity and return risk, the otherwise infinite-dimensional wealth distribution collapses to three summary statistics: aggregate capital and two covariances measuring the alignment of wealth with productivity and portfolio returns, sufficient for all contemporaneous equilibrium allocations and prices. We call the resulting object a *moment-recursive competitive equilibrium* (MRCE). Because the household problem is solved in closed form and aggregates analytically into an exact aggregate law of motion, and because the equilibrium is invariant to households' forecasts about aggregate dynamics, the framework admits an exact data inversion: it constructs the sequence of structural objects under which the observed path itself is a general-equilibrium allocation, reproducing trend, cycle, and heterogeneity at once, with heterogeneity resolved as finely as the data permit, down to the level of the individual observation.

The paper makes three contributions. First, we develop a utility form which allows closed-form policies for consumption, saving, and labor supply under incomplete markets with idiosyncratic productivity and return risk. The utility features a Stone–Geary subsistence level that scales with effective labor productivity, and an effective time endowment that increases in wealth. The individual optimal path yields a unique MRCE where $\mathcal{M} = [K, \text{cov}(z, a), \text{cov}(e, a)]$ is sufficient for all contemporaneous equilibrium allocations and prices, and the aggregate capital law of motion K' is an exact closed-form function of

$\mathcal{M}$. The saving rate is increasing and concave in wealth (Dynan et al., 2004). The framework nests several benchmarks: shutting down return heterogeneity yields Krusell–Smith; shutting down all heterogeneity recovers Brock and Mirman (1972). With return risk alone, the wealth distribution drops out as in Angeletos (2007); adding productivity heterogeneity restores its role through exactly two covariances.

Second, the equilibrium is invariant to household beliefs about aggregate dynamics. Under log utility, income and substitution effects cancel exactly, so policies depend only on current prices. This eliminates the fixed-point problem over beliefs, making the MRCE a belief-invariant recursive competitive equilibrium (BRCE). The BRCE accommodates nonstationary paths without detrending, which is essential for studying how TFP *growth*, not just TFP levels, drives inequality. Conventional approaches either compare stationary equilibria at different TFP levels or require expectations over the entire future price sequence. The BRCE sidesteps both: any TFP path can be fed into the forward simulation to obtain valid equilibrium dynamics.

Third, the closed-form structure enables an *exact general equilibrium inversion* that operates at two scales. At the aggregate scale, given U.S. output, capital, and hours (1950–2019), the aggregate equations trace the model-implied path of TFP, saving behavior, labor efficiency, prices, and distributional moments under which the observed series are themselves the equilibrium allocation, replacing the reduced-form residual accounting of Chari et al. (2007) with a structural path inside a heterogeneous-agent general equilibrium. At the household scale, taking the recovered aggregate prices as given, the same closed-form policies invert PSID and SCF micro data into individual structural parameters (z_i, e_i, β_i) wave by wave. The two scales connect, though imperfectly: the macro β_t recovered from aggregates should, in principle, equal the capital-weighted aggregate of household β_i recovered from surveys, and our two micro estimates (from PSID and SCF, two independent surveys with disjoint samples and orthogonal identification strategies) agree on this value within 1.7 pp. A residual 7–9 pp gap between the micro and macro estimates remains and is consistent with the surveys' well-known under-coverage of top-wealth households; we treat the micro-to-macro link as a validation exercise rather than an exact identity, and a formal aggregation bridge is beyond the scope of the present paper. Because the recovered objects are structural parameters in a model with explicit distributional dynamics, they support counterfactual decompositions that isolate each shock's role in inequality growth, deliver conditional forecasts that outperform a recursive AR benchmark for output and labor, and pass a Lucas-critique-style test that no household's β_i shifts around four major postwar capital-tax reforms (OBRA 1993, EGTRRA 2001, ATRA 2013, TCJA 2017). The set of reforms used for this within-survey invariance test differs from the four reforms studied in the macro tax counterfactual of Section 5.4 (Reagan ERTA 1981, OBRA 1993, EGTRRA 2001, ATRA 2013), since the Lucas-test set is restricted to reforms whose enactment is bracketed by adjacent PSID and SCF survey waves.

The inversion yields several substantive findings. A *sorting accelerator* channel links TFP growth to wealth concentration: higher TFP raises returns, which compound through near-unit-root return-wealth sorting and widen the wealth distribution via self-reinforcing feedback (Piketty, 2014). At the baseline calibration, the channel's unconditional amplification of a one-period TFP shock is modest. It becomes quantitatively decisive under persistent policy shifts: the 1981 and 2001 capital-tax reforms produce wealth-distribution divergences that widen for decades, because the near-unit-root sorting mechanism prevents mean reversion (Section 5.4). The traced primitive sequence reveals a two-dimensional business cycle. TFP is nearly orthogonal to both labor productivity and patience ($\rho(\Delta \log A, \Delta \log Z_L) = 0.142$, $\rho(\Delta \log A, \Delta\beta) = 0.105$). Labor productivity and patience, by contrast, are strongly negatively correlated ($\rho(\Delta \log Z_L, \Delta\beta) = -0.858$): labor-efficiency declines coincide with rising patience, an anti-cyclical pairing consistent with a precautionary-saving channel. The structural classification attributes individual postwar recessions to different shocks, with this labor-efficiency/patience pairing appearing as one recognizable but not universal pattern.

Related literature Bewley (1986), Huggett (1993), and Aiyagari (1994) developed the foundational heterogeneous-agent framework. Krusell and Smith (1998) showed numerically that mean wealth nearly suffices for aggregate forecasting; the MRCE identifies the omitted moments and gives exact conditions under which the approximation succeeds. The result is closely related to Den Haan and Rendahl (2010), who show that cross-sectional averages of individual-policy basis functions are natural aggregate states. Here those averages reduce to economically interpretable sorting covariances. Computational approaches to high-dimensional heterogeneous-agent dynamics include Reiter (2009), Winberry (2018), Ahn et al. (2018), and the sequence-space Jacobian method of Auclert et al. (2021). The contribution here is complementary: it gives exact nonlinear dynamics for a restricted class rather than linearized dynamics for a broad class.

The paper also builds on exact aggregation under log preferences. Brock and Mirman (1972) solve the representative-agent growth model, and Angeletos (2007) obtains aggregation with uninsured return risk. This paper shows how adding labor-productivity heterogeneity restores a role for the wealth distribution through two covariances. Bilbiie (2020) and Ravn and Sterk (2021) develop parsimonious heterogeneous-agent models with analytical aggregate dynamics; the MRCE uses a continuum of agents with rich return heterogeneity. Alvarez and Lippi (2022) develop analytical impulse responses using eigenvalue methods in menu-cost models; the moment structure here plays an analogous role for heterogeneous households.

Krueger et al. (2025) and Ando et al. (2025) achieve closed-form transition paths in a limited-commitment economy where the zero-income state reduces each type's problem to cake-eating under log utility. The MRCE takes a different tractability route from the same log-utility foundation: rather than eliminating labor income, it folds labor income into the effective return through a Stone–Geary wealth-leisure complementarity, preserving continuous idiosyncratic distributions, return heterogeneity, and endogenous labor supply. The trade-off is that the limited-commitment structure delivers a closed-form transition for K alone, while the MRCE's covariance dynamics require agent-based simulation. The two approaches are complementary: they are dual entry points into analytical heterogeneous-agent macro, each covering ground the other cannot.

The belief-invariance result is related to block-recursive equilibria (Menzio and Shi, 2010) and cognitive discounting (Gabaix, 2020), but arises endogenously from the income-substitution offset under log utility. This is the same property that Krueger et al. (2025) and Ando et al. (2025) exploit when noting that current policies are independent of future interest rates. Belief invariance is thus a shared engine across multiple tractable heterogeneous-agent structures, not a feature unique to any one market arrangement.

Belief invariance also speaks to a recent challenge to the field. Moll (2025) argues that rational expectations are the weak link in heterogeneous-agent macroeconomics: households are required to forecast the equilibrium law of motion of an infinite-dimensional distribution, an assumption that is neither computationally innocuous nor empirically supported, and survey evidence shows large, systematic belief biases that move aggregates (Bhandari et al., 2025). The MRCE offers one way to address this concern from within the rational-expectations tradition rather than against it: because the income and substitution effects of future prices cancel under log utility, optimal policies are *independent* of beliefs altogether. Whatever expectations households hold, rational, biased, or simply unspecified, the equilibrium allocations and the aggregate law of motion are unchanged. The framework therefore does not need to take a stand on expectation formation, and the inversion recovers structural shocks without assuming agents forecast the future correctly. Where belief bias does have economic content in our setting, it enters through the recovered patience parameter: the individual-specific, time-varying $\beta_{i,t}$ absorbs shifts in a household's effective valuation of the future, so a change in an agent's outlook or belief is identified as a movement in $\beta_{i,t}$ rather than as a deviation from an assumed expectation rule.

The household inversion connects to a parallel program in macro-labor that recovers latent worker and firm types from micro data. Bonhomme et al. (2019) classify latent types from matched employer–employee panels using finite mixtures; Hagedorn et al. (2017) establish nonparametric identification of sorting models from wages and labor-market transitions; and Lentz et al. (2023) and Hong et al. (2026) recover worker-type dynamics and sorting via hidden-Markov classification. Our micro inversion is methodologically distinct (it reads structural parameters off closed-form equilibrium policies rather than estimating a latent-type mixture), but it shares the same goal of recovering individual-level structural heterogeneity from observed cross-sections, and our Table 12 decomposition into permanent, life-cycle, and transitory components is a direct counterpart to the type-versus-dynamics decompositions in that literature.

The state-dependent amplification results contribute to the nonlinear business cycle literature: Storesletten et al. (2004) document countercyclical cross-sectional risk, Bloom et al. (2018) study uncertainty-driven state dependence, and Berger and Vavra (2015) analyze wealth heterogeneity in consumption dynamics. The asset pricing results relate to idiosyncratic risk and equilibrium returns (Constantinides and Duffie, 1996; Schmidt, 2016); growing evidence on heterogeneous portfolio returns (Fagereng et al., 2020; Bach et al., 2020) motivates the model's return channel.

Roadmap The remainder of the paper proceeds as follows. Section 2 sets up the model. Section 3 derives the MRCE, establishes belief invariance, and draws out its theoretical implications. Section 4 implements the inversion at two scales: aggregate (U.S. output, capital, and hours, 1950–2019) and household level (PSID and SCF), and validates both. Section 5 traces the macro and micro implications: aggregate dynamics, the sorting accelerator, conditional forecasting, tax-reform counterfactuals, household parameter distributions, and a Lucas-critique-style policy-invariance test. Section 6 concludes. Proofs, extensions, and additional applications are collected in the Online Appendix.

2 Model

A continuum of infinitely lived households face idiosyncratic labor productivity and portfolio return risk and choose consumption, saving, and labor supply each period.

2.1 Preferences

Household preferences take a Stone–Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \tag{1}$$

where c is consumption, n is labor supply, a is household wealth, z is idiosyncratic labor productivity, and $\theta > 0$ governs the relative weight on leisure. The household state $x = [a, z, e, \beta]$ and the aggregate state X are specified in Section 2.2; the wage $w(X)$, return $r(X)$, and aggregate labor productivity Z_L are determined in equilibrium and detailed in Sections 2.5–2.6 below.

The subsistence level $g(x; X) = \kappa w(X)(1 - \tau) Z_L z$ captures work-related expenses that scale with after-tax earnings.

The effective time endowment $\kappa + \xi a$ increases in wealth: $\kappa \in (0, 1)$ is the base endowment, and $\xi > 0$ captures wealth buying usable time through time-saving services (Aguiar and Hurst, 2007). This wealth-leisure complementarity links the wealth distribution to aggregate labor supply.

The three preference parameters (κ, θ, ξ) are jointly calibrated to three U.S. moments, in the spirit of Cooley and Prescott (1995) and Chang and Kim (2007). κ targets an average market-hours share near one third of the time endowment, the convention in Aiyagari–Bewley environments with endogenous labor (Chang and Kim, 2007; Heathcote et al., 2017). θ is set to deliver a Frisch elasticity at the intensive margin of approximately 0.5, the value most commonly used in macro labor calibrations (MaCurdy, 1981; Keane and Rogerson, 2012). ξ is identified from the cross-sectional gradient of hours in wealth measured by the quasi-experimental wealth-effect literature (Imbens et al., 2001; Cesarini et al., 2017); in our annual calibration the implied steady-state saving-to-labor-income ratio is 0.33, consistent with U.S. NIPA averages. The transfer share η (introduced in Section 2.6) is set so that steady-state wealth concentration reproduces a U.S. wealth Gini near 0.80 (Díaz-Giménez et al., 2011; Kuhn et al., 2020a).

2.2 States and aggregate environment

Each household's state is $x = [a, z, e, \beta]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous, with $\mathbb{E}[z] = 1$), a return-scaling factor e (exogenous, with $\mathbb{E}[e] = 1$), and the household's discount factor β (a structural patience parameter, possibly varying both across households and over time). All four display substantial cross-sectional variation (Kuhn et al., 2020b; Storesletten et al., 2004; Fagereng et al., 2020).

The aggregate state is $X = [\Phi, A, Z_L, \bar{\beta}, B, C]$: the joint cross-sectional distribution Φ over (a, z, e, β), aggregate TFP A, aggregate labor productivity Z_L, the capital-weighted aggregate discount factor $\bar{\beta} \equiv \mathbb{E}_\Phi[\beta_i a_i R_i]/\mathbb{E}_\Phi[a_i R_i]$, return dispersion B, and productivity dispersion C. The aggregate $\bar{\beta}$ is what enters the closed-form aggregate equations of Section 3 and what the macro inversion of Section 4.1 recovers period by period; the underlying individual β_i distribution is recovered wave by wave from PSID and SCF in Section 4.2. For notational simplicity, when there is no ambiguity we write β for both the individual state and its capital-weighted aggregate; the context resolves which is intended. In the quantitative application (Section 4), A, Z_L, and $\bar{\beta}$ vary at the aggregate level; B and C are held at steady state. The framework accommodates time-variation in all five (e.g., widening return dispersion during financial crises or shifting patience during demand-driven episodes). The laws of motion for the idiosyncratic shocks (z, e) are specified next; the idiosyncratic β is treated as a household-specific primitive whose cross-sectional distribution is part of Φ.

2.3 Idiosyncratic processes

The idiosyncratic productivity z and return factor e follow stationary AR(1) dynamics in levels with unit mean:

$$z' - 1 = \rho_z (z - 1) + \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0, \sigma_z^2), \tag{2}$$

$$e' - 1 = \rho_e (e - 1) + \varepsilon'_e, \quad \varepsilon'_e \sim \mathcal{N}(0, \sigma_e^2), \tag{3}$$

where σ_z, σ_e are the per-period *innovation* standard deviations (matching the calibration code), and $\varepsilon'_z, \varepsilon'_e$ are mutually independent. The unconditional variances are $\sigma_z^2/(1-\rho_z^2)$ and $\sigma_e^2/(1-\rho_e^2)$. Linearity in levels is essential: it ensures that the cross-sectional covariances $\mathrm{cov}(z, a)$ and $\mathrm{cov}(e, a)$ serve as sufficient statistics for contemporaneous equilibrium allocations (Section 3). In the quantitative analysis, each process is discretized using the Tauchen (1986) method.

Each household's effective labor productivity per hour is $Z_L z$, where Z_L is the aggregate labor productivity parameter from the state X. Z_L captures aggregate labor market conditions: utilization intensity, matching quality, and workforce composition. It enters the production function only through the efficiency-weighted labor aggregate $L \equiv Z_L \int z\, n(x; X)\, d\Phi$, introduced in Section 2.5 below; it does not appear as a separate factor multiplying $L^{1-\alpha}$. This asymmetry distinguishes Z_L from neutral TFP A: a shift in Z_L rescales effective hours one-for-one but does not change the production technology itself (see Section 4 and Online Appendix B for the structural interpretation). At steady state, $Z_L = 1$.

2.4 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x;X) = \max_{c,a',n} \quad u(c,n,a,z;X) + \beta \mathbb{E} v(x';X') \tag{4}$$

$$\text{s.t.} \quad c + a' = w(X)(1-\tau)Z_L z n + (1 + r(X)(1-\tau)e)a \tag{5}$$

$$\Phi' = \Gamma_\Phi(X) \tag{6}$$

$$a' \geq 0, \quad n \geq 0 \tag{7}$$

The household's problem does not include the transfer T: households are unaware of its existence and optimize based solely on factor income. The actual law of motion for wealth is $a'_{\text{actual}} = a'_{\text{chosen}} + T(X)$, where T is added by the government after the household's decision (see Section 2.6). The wage $w(X)$ is per hour of labor and $r(X)$ is the aggregate return on capital. The factor e scales returns: $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average. The product $Z_L z$ determines the household's effective hourly wage; Z_L captures aggregate labor market conditions while z captures individual heterogeneity. The discount factor $\beta \in (0,1)$ is an element of the aggregate state X and may vary over time, capturing shifts in patience due to financial conditions, uncertainty, or other demand-side forces. The household faces a borrowing constraint $a' \geq 0$, which is shown to be non-binding directly from the household's optimization (Corollary 1 below).

2.5 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = A K_{eff}^{\alpha} L^{1-\alpha}, \tag{8}$$

where the inputs are *efficiency-weighted* factor aggregates:

$$K_{eff} \equiv \int e\, a\, d\Phi = K + \operatorname{cov}(e,a), \qquad L \equiv Z_L \int z\, n(x;X)\, d\Phi, \tag{9}$$

with $K = \int a\, d\Phi$ denoting physical (wealth) capital and $\alpha \in (0,1)$. The return factor e_i scales the efficiency of household i's capital in production (e.g., reflecting portfolio quality or entrepreneurial ability), just as z_i scales labor productivity. Factor prices satisfy:

$$r(X) = \alpha A K_{eff}^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1-\alpha) A K_{eff}^{\alpha} L^{-\alpha}, \tag{10}$$

where δ is the depreciation rate. Competitive factor markets pay $(r+\delta)$ per efficiency unit of capital and w per efficiency unit of labor. Household i supplies $e_i a_i$ efficiency units of capital and $Z_L z_i n_i$ efficiency units of labor, earning gross capital income $(r+\delta)e_i a_i$ and labor income $wZ_L z_i n_i$ respectively. Aggregating: total capital income $(r+\delta)K_{eff} = \alpha Y$ and total labor income $wL = (1-\alpha)Y$, exhausting output. Because factor prices depend on K_{eff}/L, distributional forces (both labor supply shifts and return-wealth sorting) feed back into prices.[1]

2.6 Government policy

The government levies proportional income taxes at rate τ_K on capital income and τ_L on labor income, and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot \big[\tau_K(r+\delta)K_{eff} + \tau_L wL\big], \tag{11}$$

When $\tau_K = \tau_L = \tau$, this reduces to $T = \eta\tau Y$ (since $(r+\delta)K_{eff} + wL = Y$). After-tax prices are $\tilde{r} = r(1-\tau_K)$ and $\tilde{w} = w(1-\tau_L)$.[2] Remaining revenue finances government consumption (not in utility or production). The baseline sets $\tau_K = \tau_L = 0.30$; the Online Appendix considers time-varying rates.

The transfer plays a dual role. First, it is a fiscal policy channel: tax revenue is partially redistributed, with η controlling the degree of redistribution. Higher η compresses the wealth distribution and thins the Pareto tail. Second, it provides the additive intercept in the Kesten process governing wealth dynamics, which is necessary for a stationary distribution with a Pareto tail to exist (Kesten, 1973). Without $T > 0$, the purely multiplicative system $a' = \beta R_i a$ collapses to the degenerate distribution at zero and admits no non-degenerate

[1] Two conventions reconcile the household budget (4) with the production-side identity $(r+\delta)K_{eff} = \alpha Y$. (i) Depreciation on household i's physical capital is δa_i rather than $\delta e_i a_i$, so that aggregate depreciation $\int \delta a_i \, d\Phi = \delta K$ corresponds to the wealth-capital stock. (ii) The factor e_i is interpreted as scaling the household's *net* return r, so that effective post-depreciation capital income is $re_i a_i$. The wedge between this convention and the gross expression $(r+\delta)e_i a_i - \delta a_i$ is $\delta(e_i - 1)a_i$, which integrates to $\delta \cdot \mathrm{cov}(e, a)$ at the aggregate level (since $\mathbb{E}[e] = 1$) and is absorbed in the lump-sum transfer T, in parallel with the tax-wedge convention adopted in footnote 2 below. This preserves the linear separability $a' = \beta R_i a + T$ on which Theorem 1 depends.

[2] Government revenue in formula (11) is computed on gross capital income $(r+\delta)K_{eff}$ — consistent with U.S. corporate taxation of book profits net of depreciation — while the household's after-tax return $\tilde{r} = r(1-\tau_K)$ is defined on the net return r. The implied wedge between household tax payments and aggregate revenue is $\tau_K \delta K_{eff}$, equal to about 1.5 pp of capital income at baseline ($\tau_K = 0.30$, $\delta = 0.07$). This wedge is absorbed in the lump-sum transfer T and does not affect any household first-order condition or aggregation result; we retain the formulation because it preserves the linear separability $a' = \beta R_i a + T$ on which Theorem 1 depends.

stationary distribution.

Timing The transfer arrives after the household's consumption-saving decision: households solve their problem without knowledge of T, and the government adds T to wealth at the end of the period. This timing is essential for preserving the analytical tractability. If households anticipated T, they would incorporate its present value into effective wealth, breaking the $v = A \log a + B$ value function conjecture and the multiplicative-additive separation $a' = \beta R_i a + T$ that enables closed-form aggregation. The assumption is quantitatively innocuous: at the baseline calibration, the per-period transfer is 0.1% of mean wealth, and the present value of the entire transfer stream is approximately 2.6% of mean wealth. Incorporating this present value into effective wealth would shift saving shares by a correspondingly small amount.

2.7 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_Φ.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. *Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ.*

The difficulty is that Φ is infinite-dimensional. The next section shows that optimal policies depend on Φ only through three moments (aggregate capital and two covariances) that serve as sufficient statistics for all contemporaneous equilibrium allocations.

3 Moment-Recursive Competitive Equilibrium

3.1 Analytical household policies

The household's optimal policies have explicit solutions that reduce the infinite-dimensional distribution to a finite moment vector.

Proposition 1 (Analytical household policies and wealth dynamics)**.**
The optimal consumption and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1+r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)Z_L z \tag{12}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)Z_L z},\ 0\right\} \tag{13}$$

The household's chosen saving is $\hat{a}'(x;X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right)$. *The actual law of motion for wealth, after the unanticipated government transfer, is:*

$$a'(x;X) = \hat{a}'(x;X) + T(X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) + T(X). \tag{14}$$

Proof. The budget constraint can be rewritten as:

$$(1+\theta)(c - \kappa Z_L z w(X)(1-\tau)) + a' = \left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) a. \tag{15}$$

Under log utility, the standard result of Brock and Mirman (1972) yields $a' = \beta a R_i$. The government then adds the unanticipated transfer: $a'_{\text{actual}} = \beta a R_i + T(X)$. ■

Consumption is linear in wealth with source-dependent MPCs. The saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component (enabling aggregation) and an additive transfer (ensuring stationarity).

3.2 The saving rate

The saving rate has a closed-form characterization.

Proposition 2 (Analytical saving rate)**.**
The saving rate $s(x;X) = \frac{a'(x;X)-a}{y(x;X)}$, *where* $y(x;X) = w(X)(1-\tau)Z_L z n(x;X) + r(X)(1-\tau)ea + T(X)$ *is after-tax income plus transfer, satisfies:*

$$s(x;X) = \frac{a[\beta(1+\tilde{r}\,e+\tilde{w}\,Z_L z\xi)-1]+T}{\tilde{w}\,Z_L z\,\kappa\theta + a\left[\frac{\tilde{w}\,Z_L z\,\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+\tilde{r}\,e)}{1+\theta} + \tilde{r}\,e\right] + T}, \tag{16}$$

where $\tilde{r} = r(X)(1-\tau)$ *and* $\tilde{w} = w(X)(1-\tau)$, *with the following properties:*

(i) $\lim_{a\to\infty} s(x;X) = \dfrac{\beta(1+\tilde{r}\,e+\tilde{w}\,Z_L z\xi)-1}{\frac{\tilde{w}\,Z_L z\,\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+\tilde{r}\,e)}{1+\theta} + \tilde{r}\,e}$

(ii) $\lim_{a\to 0} s(x;X) > 0$ *due to the transfer* $T(X) > 0$

(iii) $\frac{\partial s}{\partial a} > 0$ *and* $\frac{\partial^2 s}{\partial a^2} < 0$.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive at zero wealth (due to T), increasing, and concave, matching the pattern documented by Dynan et al. (2004). Concavity generates mean reversion in relative wealth: as wealth rises, the subsistence floor absorbs a diminishing share and the saving rate converges to a constant.

Non-binding borrowing constraint An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint)**.**
The household's chosen saving satisfies $\hat{a}'(x;X) = \beta a R_i(x;X) \geq 0$ *for every* (a, z, e, X) *with* $a \geq 0$*, with equality only on the measure-zero set* $\{a = 0\}$*. The constraint* $a' \geq 0$ *is therefore slack in the household's optimization problem, independent of the transfer* T*.*

Proof. By Proposition 1, $\hat{a}'(x;X) = \beta a R_i$ with $R_i = 1+\tilde{r}(X)\, e+\tilde{w}(X)\, Z_L\, z\, \xi$. The labor-side term $\tilde{w} Z_L z \xi \geq 0$, and the capital-side term $1 + \tilde{r}e \geq 0$ whenever the after-tax return does not wipe out more than 100% of capital in one period, a mild condition satisfied throughout our calibration. With $\beta \in (0, 1)$ and $a \geq 0$, the product is non-negative. The transfer T does not enter the household's Kuhn–Tucker condition because households are unaware of T when choosing a'; it is added after the decision and only further raises the realized a'. ■

Slackness in the optimization problem is the structural property that keeps the policy globally linear and enables the analytical aggregation of Theorem 1. It rests on log preferences, which give a constant saving share β out of effective wealth aR_i, and on non-negativity of the effective return, not on the size of fiscal transfers.

3.3 Existence and characterization

Definition 2 (Moment-recursive competitive equilibrium (MRCE))**.**
A collection $(g_c, g_a, g_n, v, \mathit{\Gamma}_M, K, L, w, r, \mathcal{M})$ *is a moment-recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given* $\mathit{\Gamma}_M$*.*

2. K, L *satisfy the production sector's optimality conditions.*

3. *Factor markets clear.*

4. Γ_M satisfies $\Gamma_M(\mathcal{M}(\Phi), A, Z_L, \beta) = \mathcal{M}(\Phi')$, where $\mathcal{M}$ extracts the sufficient moments from Φ, and Φ' is the distribution of individual states induced by g_a and the exogenous processes.

The key departure from the RCE is that Γ_M maps a *finite* moment vector to contemporaneous equilibrium allocations and prices. The sufficient statistics are aggregate capital K and two covariances: $\text{cov}(z, a)$ measures the alignment between earnings ability and wealth; $\text{cov}(e, a)$ measures the alignment between portfolio returns and wealth.

Proposition 3 (Unique existence of the MRCE)**.**
The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a), cov(e, a)]$. Define after-tax prices $\tilde{r} = r(\mathcal{M}, A)(1 - \tau)$, $\tilde{w} = w(\mathcal{M}, A)(1 - \tau)$, and the transfer $T = \eta \cdot Y \cdot \tau$. Then:

(i) Contemporaneous sufficiency. Given $\mathcal{M}$ and the aggregate state (A, Z_L, β), all equilibrium allocations (individual consumption $c(x; X)$, saving $a'(x; X)$, labor supply $n(x; X)$, and aggregate prices r, w, L) are fully determined without reference to the distribution Φ beyond its moments $\mathcal{M}$.

(ii) Aggregate capital law of motion. K' has an exact closed-form expression:

$$K' = \beta \left[(1 + \tilde{r} + \tilde{w} Z_L \xi) K + \tilde{w} Z_L \xi \; cov(z, a) + \tilde{r} \; cov(e, a)\right] + T \tag{17}$$

(iii) Factor prices and aggregate labor. Defining $K_{eff} = K + cov(e, a)$:

$$r(\mathcal{M}, A) = \alpha A \left(\frac{L}{K_{eff}}\right)^{1-\alpha} - \delta \tag{18}$$

$$w(\mathcal{M}, A) = (1 - \alpha) A \left(\frac{K_{eff}}{L}\right)^{\alpha} \tag{19}$$

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L \kappa + \frac{Z_L \xi (1 + \theta\beta)(K + cov(z, a))}{1 + \theta} - \frac{\theta(1 - \beta)}{(1 + \theta)\tilde{w}} [K(1 + \tilde{r}) + cov(e, a)\tilde{r}] \tag{20}$$

Proof. See Appendix A.3 of the Online Appendix. ■

The K' equation reveals three distinct channels. The first term $(1 + \tilde{r} + \tilde{w} Z_L \xi) K$ is the representative-agent channel. The second, $\tilde{r}\,\text{cov}(e, a)$, is the *return-sorting channel*: high-return households are disproportionately wealthy, raising aggregate saving above what average returns and average wealth would predict. The third, $\tilde{w} Z_L \xi\,\text{cov}(z, a)$, is the *productivity-sorting channel*: productive-wealthy households generate additional saving through wealth-leisure complementarity. These terms make dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional dynamics and the closure problem While $\mathcal{M}$ is sufficient for all contemporaneous equilibrium allocations and K' is exactly determined, the forward evolution of $\operatorname{cov}(z', a')$ and $\operatorname{cov}(e', a')$ generally depends on higher-order moments of the joint distribution Φ. Specifically, from $a' = \beta a\big((1 + \tilde{r}e) + \tilde{w} Z_L z\xi\big) + T$ and $z' = \rho_z z + \varepsilon_z$:

$$\operatorname{cov}(z', a') = \rho_z \beta \big[(1 + \tilde{r} + \tilde{w} Z_L \xi) \operatorname{cov}(z, a) + \tilde{w} Z_L \xi\, \mathbb{E}[\hat{z}^2 a] + \tilde{r}\, \mathbb{E}[\hat{z}\hat{e}a]\big], \tag{21}$$

where $\hat{z} = z - 1$ and $\hat{e} = e - 1$. The terms $\mathbb{E}[\hat{z}^2 a]$ and $\mathbb{E}[\hat{z}\hat{e}a]$ involve three-way cross-moments that are not functions of $\mathcal{M}$ alone. Under joint normality of $(\hat{z}, \hat{e}, a)$, these reduce to $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$ and $\mathbb{E}[\hat{z}\hat{e}a] = 0$ (by the Isserlis/Wick theorem, using independence of z and e), and the moment dynamics close. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality.

Therefore, tracking the full distribution via agent-based simulation is necessary for the forward evolution of distributional moments, while the exact household policies, the K' equation (17), and all contemporaneous allocations are preserved analytically. The quantitative analysis in Section 4 employs this approach: exact policies govern each agent's behavior, and the distributional moments are computed from the simulated cross-section.

3.4 Ergodicity

Corollary 2 (Kesten stationarity)**.**
The individual law of motion $a' = \beta a\big((1 + \tilde{r}e) + \tilde{w} Z_L z\xi\big) + T$ is a Kesten process with multiplicative coefficient $R_i = \beta\big((1 + \tilde{r}e) + \tilde{w} Z_L z\xi\big)$ and additive intercept $T > 0$. When $\mathbb{E}[\log R_i] < 0$, the wealth distribution admits a unique stationary solution with Pareto tails (Kesten, 1973). The additive transfer $T > 0$ is essential: without it, the purely multiplicative system admits only the degenerate distribution at zero.

Proof. See Appendix A.4 of the Online Appendix. ■

The near-unit-root property of the dominant eigenvalue (empirically ≈ 1.01 in the calibrated model) generates slow mean reversion of the wealth distribution and amplifies shock propagation, as shown in Section 4.

3.5 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue)**.**
Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a)]$, and the aggregate capital law of motion is:

$$K' = \beta \left[(1 + \tilde{r} + \tilde{w} Z_L \xi) K + \tilde{w} Z_L \xi \; cov(z, a)\right] + T \tag{22}$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \mathrm{cov}(z, a)]$, the analytical counterpart of Krusell and Smith (1998). Capital alone suffices if and only if $\mathrm{cov}(z, a) = 0$. Approximate aggregation holds when $\mathrm{cov}(z, a)$ is small relative to K, i.e., when productivity shocks are weakly persistent or wealth mobility is high. The MRCE thus provides exact conditions under which the Krusell and Smith (1998) finding holds as a theorem. Further discussion appears in Appendix B.3.

Corollary 4 (Representative-agent analogue)**.**
Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta \left[(1 + \tilde{r} + \tilde{w} Z_L \xi) K\right] + T, \tag{23}$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A, Z_L, \beta) = \frac{1 - \beta}{1 + \theta}(1 + \tilde{r} + \tilde{w} Z_L \xi) K + \kappa \tilde{w} Z_L. \tag{24}$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mirman, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

3.6 Belief invariance

A distinctive feature of the MRCE is invariance to household beliefs about aggregate dynamics, eliminating the fixed-point problem that dominates standard heterogeneous-agent computation.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE)).
A collection $(g_c, g_a, g_n, v, \mathit{\Gamma}_\Phi, K, L, w, r)$ is a belief-invariant recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. *$\mathit{\Gamma}_\Phi$ satisfies $\mathit{\Gamma}_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the policy function g_a and the exogenous idiosyncratic process to Φ.*

Theorem 1 (Belief invariance).
Fix the conjecture that the current discount factor $\bar{\beta}_t$ is treated as permanent (the constant-β continuation belief). In the BRCE, the resulting law of motion $\mathit{\Gamma}_\Phi$ is then invariant to households' expectations about future aggregate prices and quantities $(A_{t+s}, Z_{L,t+s}, \bar{\beta}_{t+s}, \ldots)$ for any $s \geq 1$. That is, given the current discount wedge, no further assumption about the path of future aggregates is required to pin down equilibrium policies.

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x; X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi) + T(X)$ depends only on current prices and Z_L, not on forecasts. Under log utility, income and substitution effects of future return changes exactly offset, so all policies are belief-invariant and the aggregate law of motion inherits this property.

Proposition 4 (Unique existence of the BRCE).
The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

Because the MRCE and BRCE coincide, the equilibrium is computed by forward simulation from any initial distribution Φ_0 and aggregate sequences $\{A_t, Z_{L,t}, \beta_t\}$. No fixed-point iteration is required, no assumption on expectations need be made, and no stationarity is imposed. Each period, the exact household policies determine individual decisions, aggregate moments are computed from the cross-section, and the distribution evolves forward. The same procedure applies whether aggregates follow stationary processes, deterministic trends, or both. This also enables the exact inversion: structural shocks can be recovered from data without taking a stand on expectation formation.

Corollary 5 (No terminal condition)**.**
The equilibrium path $\{\Phi_t, K_t, L_t, w_t, r_t\}$ generated by the forward simulation is fully determined by the initial distribution Φ_0 and the aggregate sequence $\{A_t, Z_{L,t}, \beta_t\}$. It does not depend on a terminal date, a terminal value, or a transversality condition.

Proof. By Theorem 1, the household saving policy $a'(x; X) = \beta a R_i + T$ depends only on the current state (a, z, e, X), with no expectation of future X'. Given Φ_t and $X_t = (A_t, Z_{L,t}, \beta_t)$, prices (r_t, w_t) are determined by market clearing, and Φ_{t+1} is obtained by applying the policy household-by-household. Iterating from Φ_0 along the exogenous sequence $\{X_t\}$ pins down all equilibrium paths as a first-order forward recursion. No terminal date, terminal value, or transversality condition appears in the construction. ■

This is a direct consequence of belief invariance and warrants emphasis, because it marks a sharp departure from forward-looking equilibrium concepts. In a standard recursive competitive equilibrium, individual policies solve an intertemporal optimization whose Euler equation pins down behavior only together with a transversality condition; the equilibrium path is the solution to a two-point boundary-value problem, anchored at the initial distribution and at a terminal (or stationary) endpoint. The endpoint is not incidental: it is what selects the saddle path and rules out explosive or self-fulfilling trajectories. Conventional solution methods therefore require either a terminal steady state (shooting and MIT-shock methods) or an infinite horizon over which expectations of the entire future price sequence are formed.

The MRCE has no such requirement. Under log utility the income and substitution effects of future prices cancel, so the optimal saving share is the constant β_t regardless of what follows; the transversality condition holds automatically and imposes no additional restriction on the path. The equilibrium is the solution to an *initial-value* problem, integrated forward like a law of motion in the natural sciences, rather than a boundary-value problem that must be anchored at both ends. Practically, this is what makes the inversion well posed: the data need not run to a terminal steady state, the sample can end mid-transition in 2019, and no assumption about the post-sample future is required to rationalize the observed path. The endpoint simply does not enter.

Trend, cycle, and the raw-data inversion. The same property gives the framework an empirical advantage that is otherwise hard to come by. Conventional structural macro models are defined relative to a stationary equilibrium or a deterministic trend that pins down the long-run path; confronting them with data therefore requires pre-treatment in which the raw series is detrended, HP-filtered, or otherwise reduced to a cycle-only object, while trend growth, seasonality, and structural breaks are stripped out before estimation and layered back in through ad hoc reduced-form devices. The MRCE inverts the original aggregates as they are. Trend growth and business-cycle fluctuations are jointly absorbed by the recovered shock sequence $(A_t, \beta_t, Z_{L,t})$ within a single equilibrium framework, with no separate treatment of the long run and the cycle and no auxiliary assumption that the economy ever reaches a stationary distribution. The recovered A_t carries the postwar growth trend together with its cyclical movements; β_t and $Z_{L,t}$ trace persistent and transient deviations within the same equation system. This is why the inversion runs directly on the raw U.S. 1950–2019 series and reproduces them period by period: trend, cycle, and distributional drift are equilibrium properties of the same closed-form system, not preprocessing choices imposed before the model is solved.

Remark 1 (Saving share under time-varying β and the constant-β conjecture)**.** *Since $\bar\beta$ is an element of the aggregate state X, it may vary over time. With time-varying $\bar\beta$, the value function takes the form $v(a,z,e;X) = \zeta \log a + B(z,e,X)$, where ζ satisfies $\zeta = (1+\theta) + \bar\beta\,\mathbb{E}[\zeta']$. The closed-form $\zeta = (1+\theta)/(1-\bar\beta)$ and saving share $s = \bar\beta$ obtain under the conjecture that households treat the current $\bar\beta_t$ as permanent. This conjecture is the only* continuation-belief *restriction the framework imposes; given it, belief invariance (Theorem 1) then renders policies independent of expectations about all other aggregate paths $(A_{t+s}, Z_{L,t+s}, \ldots)$ for $s \geq 1$. The Online Appendix A.8 reports the corresponding ζ recursion under alternative belief specifications about $\bar\beta$, and shows that the equilibrium covariance and capital-law-of-motion expressions are continuous in the conjecture, with the constant-$\bar\beta$ case as the natural focal point. Under the maintained conjecture, the saving policy is $a'(x;X) = \bar\beta\, a\big((1+\tilde r e) + \tilde w Z_L \xi z\big) + T$ and the aggregate capital law of motion is $K' = \bar\beta\big[(1+\tilde r + \tilde w Z_L\xi)K + \tilde w Z_L \xi\, cov(z,a) + \tilde r\, cov(e,a)\big] + T$.*

Reconciling with the expectations literature. A long tradition in macroeconomics studies how variation in beliefs about the future moves equilibrium quantities. The classic Bellman decomposition

$$v(x;X) \;=\; u(x;X) \;+\; \beta\,\mathbb{E}\big[\,v(x';X') \,\big|\, x, X\,\big] \tag{25}$$

isolates a forward-looking term $\beta\,\mathbb{E}[v(x';X')]$ whose movements can be attributed either to changes in patience β or to changes in the expected continuation value $\mathbb{E}[v(x';X')]$. When β is allowed to vary, over time as cyclical patience or across households as preference heterogeneity, this attribution is fundamentally non-identified from current behavior alone: a shift in the forward-looking term is consistent with either source. The expectations literature breaks the tie by holding β fixed and reading off variation in $\mathbb{E}[v]$, either inferentially or through direct survey measurement of subjective beliefs (Moll, 2025; Bhandari et al., 2025).

The MRCE breaks the tie the opposite way. Belief invariance under log utility makes $\mathbb{E}[v(x'; X')]$ irrelevant to optimal policies (Theorem 1); only the current β enters the saving rule. The current β is then flexibly identified at both scales: time-varying β_t from the aggregate inversion of Section 4.1 and cross-sectional β_i from PSID and SCF in Section 4.2, with no auxiliary belief assumption. Whatever the expectations literature attributes to belief shifts, the MRCE attributes to patience variation; the two interpretations are observationally equivalent at the level of policies, so the identification of the structural object loads onto β. The recovered β_t trajectory and the β_i distribution therefore carry the same forward-looking content that other approaches recover from $\mathbb{E}[v]$, but expressed as a directly identified, model-consistent patience parameter. The strong negative correlation $\rho(\Delta \log Z_{L,t}, \Delta\beta_t) = -0.858$ reported in Section 4, in which aggregate patience rises sharply when labor productivity falls, is then naturally read as the macro-level signature of a precautionary expectations channel; the wave-to-wave dispersion of β_i documented in Section 5.5 is its cross-sectional analogue.

3.7 Macroeconomic implications

State-dependent aggregate dynamics The capital law of motion (17) decomposes into three channels: the representative-agent channel $(1+\tilde{r}+\tilde{w}Z_L\xi)K$, the return-sorting channel $\tilde{r}\operatorname{cov}(e,a)$, and the productivity-sorting channel $\tilde{w}Z_L\xi\operatorname{cov}(z,a)$. These distributional terms make aggregate dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional channels The aggregate consumption function decomposes as:

$$C(\mathcal{M}, A, Z_L, \beta) = \underbrace{\frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L}_{\text{Representative-agent level}} + \underbrace{\frac{1-\beta}{1+\theta}\left(\tilde{r}\operatorname{cov}(e,a) + \tilde{w}Z_L\xi\operatorname{cov}(z,a)\right)}_{\text{Distributional channel}} \tag{26}$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L\kappa + \underbrace{\frac{Z_L\xi(1+\theta\beta)(K+\operatorname{cov}(z,a))}{1+\theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r}) + \tilde{r}\overbrace{\operatorname{cov}(e,a)}^{\text{Dist. channel}}]}{(1+\theta)\tilde{w}}}_{\text{Wealth effect}} \tag{27}$$

The wealth-effect term is amplified by $\operatorname{cov}(e,a)$: wealthy households earning high returns consume more leisure, reducing labor supply. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response as the households with the largest wealth gains withdraw the most labor.

State-dependent inequality dynamics The K' equation (17) generates a "wealth spiral": productivity shocks raise capital, which strengthens return-wealth sorting through $\text{cov}(e, a)$, which raises future capital. Discounting ($\beta < 1$) and the transfer T counterbalance. The persistence of the sorting mechanism, governed by the near-unit-root dynamics of the wealth distribution, controls the speed of concentration (Appendix B.4).

3.8 Asset pricing implications

Individual return heterogeneity, traditionally viewed as diversifiable, affects aggregate prices once it is sorted with wealth (Constantinides and Duffie, 1996; Storesletten et al., 2004). Households with persistently high returns accumulate disproportionate wealth, shifting labor supply and equilibrium returns:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \tag{28}$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

3.9 Why forward-solving cannot do this

Three properties make the inversion possible: contemporaneous sufficiency, belief invariance, and the absence of a terminal condition. The forward-solving paradigm has none of them. The resulting contrast is the methodological core of the paper. Its operational consequence is that the model's baseline is the actual historical economy rather than a calibrated approximation of it: interventions, forecasts, and decompositions are measured against a path that already reproduces the data exactly, not against a stylized stand-in.

The model is individual-based, not state-space-based. Conventional heterogeneous-agent methods approximate the equilibrium as a function of a low-dimensional aggregate state: a finite set of distributional moments (Krusell and Smith, 1998), polynomial or spline coefficients of the distribution (Reiter, 2009; Winberry, 2018), or a linearization around the stationary distribution (Ahn et al., 2018; Auclert et al., 2021). The object solved for is a policy function on a truncated state space; accuracy is bounded by how well the truncation captures the true distribution. The MRCE does not approximate. Each household carries its own (a_i, z_i, e_i, β_i) and follows the exact closed-form policy; aggregates are computed from the realized cross-section, not from a projection onto basis functions. Every agent in the simulation, and every household in the survey inversion, is a distinct structural unit rather than a grid point or a moment. The three sufficient statistics $\mathcal{M}$ are an exact *property* of the system, not an approximating *choice* imposed on it.

Forward-solving runs the wrong direction for inversion. Forward-solving takes parameters and shock processes as inputs and produces simulated data as output. Going the other way, from data to shocks, requires an outer estimation loop that re-solves the model for every parameter guess. This is feasible for a handful of aggregate parameters; it cannot recover a period-by-period sequence of structural shocks that rationalizes the data *exactly*, and it cannot scale to tens of thousands of households wave by wave. The MRCE reverses this comparison. Rather than simulating artificial data and measuring the gap to history, the equilibrium restrictions are imposed directly on the historical path: contemporaneous allocations are closed-form functions of $(\mathcal{M}, A_t, Z_{L,t}, \bar{\beta}_t)$, so the path itself pins down the supporting sequence of primitives and distributional states, with no outer estimation loop and no re-solving.

Expectations and the terminal condition block the inversion in forward-looking models. Even setting computation aside, a forward-looking equilibrium is not invertible in principle. Current policies depend on expected future prices, so the data at t are model-consistent only relative to a posited belief rule and a terminal anchor (Corollary 5). Two different belief specifications rationalize the same t with two different shocks: the structural object is not identified without committing to one. This is the rational-expectations bottleneck that Moll (2025) identifies. Belief invariance dissolves both halves of it. Under log utility, policies are independent of expectations and the equilibrium is an initial-value problem, so the observed path pins down its supporting primitive sequence without any auxiliary assumption about how agents forecast or where the path ends.

Counterfactuals are built on a sharply matched model. Because the inversion fits output, capital, and hours period by period with no residual, the counterfactual baseline is the actual historical economy rather than a model that matches selected moments on average. When we shut down a shock or alter a tax rate (Section 5.4), the comparison is against a path that reproduced the data exactly, so the counterfactual difference is attributable to the intervention rather than to baseline misfit. A forward-solved model calibrated to long-run moments cannot make this claim: its baseline already departs from the data, and counterfactual differences are confounded with that gap. Sharp historical matching is not cosmetic; it is what gives the counterfactuals their interpretation.

4 Structural inversion at two scales

The analytical structure of Section 3 admits an exact inversion at two scales. At the aggregate scale, U.S. time series on output, capital, and hours are treated as the path to be traced; the aggregate equations of Section 3 pin down the corresponding period-by-period sequence of aggregate primitives $(A_t, \bar{\beta}_t, Z_{L,t})$, prices, and distributional states. At the household scale, taking the macro-inverted prices as given, PSID and SCF micro data analogously pin down the individual structural parameters (z_i, e_i, β_i) wave by wave. The two scales connect, though imperfectly: the macro β_t traced from aggregates should, in principle, equal the capital-weighted aggregate of household β_i traced from surveys, but a residual gap remains; we treat the link as a validation exercise rather than an exact identity (Section 5.7). This section sets up both inversions and reports the recovery diagnostics. The substantive findings driven by the recovered objects are then collected in Section 5.

4.1 Aggregate inversion

The application uses annual U.S. data from 1950 to 2019 on real per-capita GDP, the per-capita capital stock, and aggregate hours. The aggregate equations are the production function, labor supply, and capital accumulation:

$$Y_t = A_t K_{eff,t}^{\alpha} L_t^{1-\alpha}, \tag{29}$$

$$L_t = Z_{L,t}\kappa + \frac{Z_{L,t}\,\tilde{\xi}_t(1+\theta\beta_t)}{1+\theta}\left(K_t + \mathrm{cov}_t(z,a)\right) - \frac{\theta(1-\beta_t)}{(1+\theta)\tilde{w}_t}\big(K_t(1+\tilde{r}_t) + \tilde{r}_t\,\mathrm{cov}_t(e,a)\big), \tag{30}$$

$$K_{t+1} = \beta_t\left[(1+\tilde{r}_t+\tilde{w}_t Z_{L,t}\tilde{\xi}_t)K_t + \tilde{r}_t\,\mathrm{cov}_t(e,a) + \tilde{w}_t Z_{L,t}\tilde{\xi}_t\,\mathrm{cov}_t(z,a)\right] + T_t, \tag{31}$$

where $\tilde{\xi}_t \equiv \xi/A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient (Online Appendix B). Note that equation (30) is implicit in L because prices depend on L through the marginal products (Online Appendix A proves unique existence). The distributional covariances entering $K_{eff,t}$ are obtained from a preliminary two-series inversion that provides the moment-recursion covariance path (Online Appendix E). Given the distributional state, these equations pin down $(A_t, \beta_t, Z_{L,t})$ period by period along the observed path. Production pins down A_t, capital accumulation pins down the saving-margin component β_t, and labor supply pins down the labor-efficiency component $Z_{L,t}$, each as a structural primitive of the household-firm equilibrium rather than as a reduced-form residual. The distribution is then simulated forward using the exact household policy.

The recovered shocks have distinct structural interpretations. A_t captures neutral total factor productivity. β_t captures time-varying patience, or equivalently demand-side forces operating through the saving margin. $Z_{L,t}$ is a model primitive (Section 2), not a reduced-form residual. Three features distinguish it from the labor wedge of Chari et al. (2007). First, it has a direct structural interpretation: time variation maps to shifts in labor force composition, matching quality, or utilization intensity. Second, $Z_{L,t}$ scales household decisions but not the production function, making it separately identified from TFP. Third, it has distributional consequences: $Z_{L,t}$ enters the saving policy and equilibrium prices, so shifts in labor productivity alter the dynamics of wealth inequality.

The calibration is deliberately conventional. The capital share α_t and depreciation rate δ_t are time-varying, drawn from Penn World Tables and BEA fixed-assets data respectively. The analytical expressions carry over with α replaced by α_t period by period, since each period's equilibrium is contemporaneous. The reference annual discount factor $\beta = 0.96$ is the standard value (Cooley and Prescott, 1995). The leisure weight $\theta = 4$ delivers an intensive-margin Frisch elasticity of approximately 0.5 at the calibrated average-hours share, in line with the macro labor literature (MaCurdy, 1981; Keane and Rogerson, 2012; Chang and Kim, 2007; Heathcote et al., 2017); Online Appendix E.12 reports robustness. Idiosyncratic earnings risk follows Storesletten et al. (2004), and idiosyncratic return risk follows Fagereng et al. (2020). The transfer share η is set so steady-state wealth concentration matches a U.S. wealth Gini near 0.80 (Díaz-Giménez et al., 2011; Kuhn et al., 2020a).

4.2 Household inversion

The same MRCE policies can be inverted at the household level. This subsection runs the inversion on PSID and the Survey of Consumer Finances (SCF), treating each survey as a sub-population that takes the macro-inverted GE prices as exogenous. The exercise identifies structural parameters (z_i, e_i, β_i) at the household level, provides an external validation of the macro β_t as a capital-weighted aggregate of micro β_i, and sharpens the distributional content of the framework with direct cross-sectional evidence on sorting.

PSID and SCF as sub-populations. A key methodological point is that aggregate prices (r_t, w_t), the labor productivity index $Z_{L,t}$, the transfer T_t, and the growth-adjusted

Table 1: Calibrated parameters

Parameters	Description	Value	Source
Data-anchored (time-varying)			
α_t	Capital share	0.351–0.414	Penn World Tables
δ_t	Depreciation rate	0.037–0.058	BEA Fixed Assets
A_t	Aggregate TFP	Inverted	GDP, capital, hours
β_t	Discount factor	Inverted	Capital accumulation
$Z_{L,t}$	Labor productivity	Inverted	Labor supply
Calibrated (fixed)			
β	Reference discount factor	0.9600	Annual standard
θ	Labor disutility	4.0000	Frisch elasticity ≈ 0.5
κ	Common time endowment	0.2700	Avg. market hours share
ξ	Wealth–leisure complementarity	0.0213	Wealth-effect gradient on hours
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Idiosyncratic risk			
ρ_z	Prod. persistence	0.9500	Storesletten et al. (2004)
σ_z	Prod. volatility	0.1200	Storesletten et al. (2004)
ρ_e	Return persistence	0.9990	Fagereng et al. (2020)
σ_e	Return volatility	0.0100	Fagereng et al. (2020)

Notes: Calibration is annual. The table reports the baseline used in the U.S. inversion and in the distributional simulation.

complementarity $\tilde{\xi}_t$ are taken from the macro inversion, not re-estimated from the household sample. PSID and SCF are not the universe: PSID's wealth coverage excludes the very rich; SCF over-samples the wealthy through the list-sample but is cross-sectional. Re-estimating prices from a non-representative sample biases identification. Under the universe-level GE prices, each surveyed household solves the MRCE problem of Section 2, and the closed-form policies of Proposition 1 deliver direct measurement equations.

Measurement equations. For each household i in wave t, three observables identify three structural parameters in closed form, with no consumption data required and no numerical iteration.

Productivity z_i from the FTE wage. The model implies $W_i^d = \tilde{w}_t Z_{L,t} z_i$ where W_i^d is the after-tax full-time-equivalent wage. Pre-tax both sides cancel the labor tax, giving $z_i = W_{\text{nom},i}/(w_t Z_{L,t})$. We renormalize z_i within wave so that $E_w[z_i \mid \text{worker}] = 1$, absorbing the universe-vs-sample wage-level gap that arises because the macro w_t averages over the total population while the survey conditions on workers.

Return factor e_i from capital income. The model implies $\mathrm{KI}_i = r_t e_i a_i$ in pre-tax terms (or $\tilde{r}_t e_i a_i$ post-tax), so $e_i \propto \mathrm{KI}_i/(r_t a_i)$. We compute $e_i^{\mathrm{raw}} = \mathrm{KI}_i/(r_t a_i)$ and renormalize to $E_w[e_i \mid \mathrm{KI} > 0] = 1$ within wave; the renormalization absorbs both the gross-vs-net-tax convention $(1-\tau_K)$ and any sample-level reporting biases. Households reporting zero capital income (typically those with wealth concentrated in housing or retirement accounts that do not generate reported interest or dividends) have e_i imputed to 1; the imputation rate is 50–65% in PSID and 30–55% in SCF.

Discount factor β_i. PSID is biennial. Iterating the saving policy $a_{t+1} = \beta a_t R_t + T_t$ over two years and assuming within-window constant prices yields $a_{t+2} = \beta_i^2 a_{i,t} R_i^2 + T_t(1 + \beta_i R_i)$, where $R_i = 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i$. This is a quadratic in β_i with closed-form positive root

$$\beta_i^{\mathrm{PSID}} = \frac{-T_t + \sqrt{T_t^2 + 4a_{i,t}(a_{i,t+2} - T_t)}}{2a_{i,t} R_i}. \tag{32}$$

SCF is cross-sectional. The intratemporal labor FOC combined with the consumption equation gives

$$\beta_i^{\mathrm{SCF}} = 1 - \frac{(1+\theta)\,\tilde{w}_t Z_{L,t}\, z_i\,(\kappa + \tilde{\xi}_t a_i - n_i)}{\theta R_i a_i}, \tag{33}$$

where n_i is the household's labor share. Both formulas are exact closed forms in the inverted parameters; the θ does not enter (32), and consumption does not enter either.

4.3 Recovery and validation

We now validate both inversions: the aggregate inversion against Krusell–Smith simulated benchmarks and against external labor and saving indicators not used in identification, and the household inversion against cross-survey replication.

Krusell–Smith validation of the aggregate inversion. We begin the aggregate diagnostics with a falsification test: whether the inversion spuriously creates additional shocks when the data are generated by a model with only aggregate TFP risk. We simulate three Krusell–Smith variants and apply the same inversion to simulated output, capital, and hours. Figure 1 and Table 2 show that the recovered TFP matches the true process, while β_t and $Z_{L,t}$ remain essentially constant. Two extensions, one with endogenous labor supply and one with Frisch labor supply and convex adjustment costs, deliver the same result. The inversion therefore does not mechanically manufacture preference or labor shocks when the true data-generating process has only aggregate TFP.

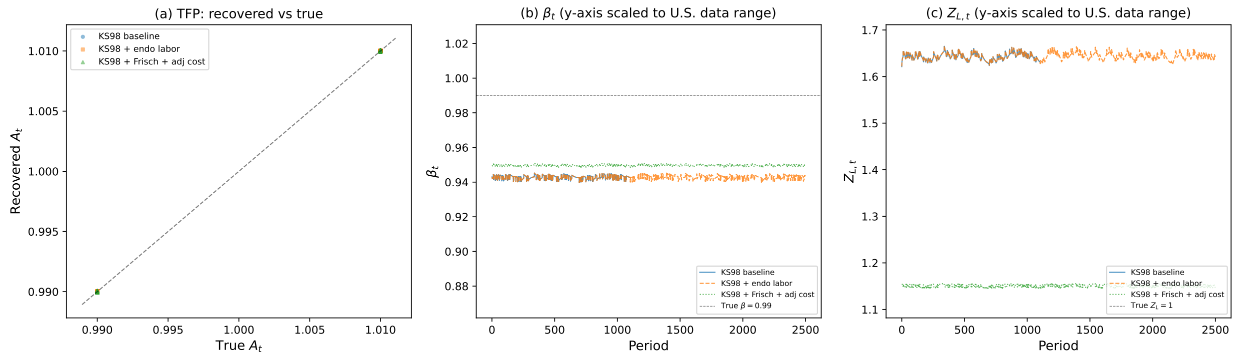

Figure 1: Krusell–Smith validation: recovered shocks from simulated data

Notes: Sequential inversion applied to simulated (Y_t, K_t, L_t) from three Krusell–Smith variants. The recovered TFP aligns with the true aggregate TFP process, while the recovered β_t and $Z_{L,t}$ series are nearly flat on the U.S. data scale.

Applying the same inversion to U.S. data reveals economically significant departures from the Krusell–Smith benchmark. The discount factor β_t fluctuates more than in the simulated economies, but the labor-efficiency wedge is the dominant departure. U.S. capital accumulation is well approximated by a single TFP shock with near-constant preferences, while labor-market fluctuations require an additional structural margin that a one-shock Krusell–Smith economy cannot generate.

Table 2: Krusell–Smith validation: inversion accuracy

	mean$\lvert A^{\text{inv}} - A^{\text{true}}\rvert/A$	CV(β)	$\sigma(\Delta \log Z_L)$	mean$\lvert Y^{\text{inv}} - Y^{\text{true}}\rvert/Y$
KS98 baseline	7.5×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + endo labor	7.3×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + Frisch + adj cost	2.5×10^{-5}	0.05%	0.18%	$< 10^{-15}$
U.S. data (1950–2019)	—	1.78%	3.42%	exact

Notes: In the simulated economies, β and Z_L are constant by construction and only aggregate TFP varies. The final row applies the same inversion to U.S. data.

Recovered series Figure 2 reports the recovered structural series. TFP grows from 0.807 in 1950 to 1.79 in 2019, tracing the postwar expansion, the 1970s slowdown, the IT revolution, and post-2008 stagnation. The saving-margin wedge β_t is persistent and countercyclical, with

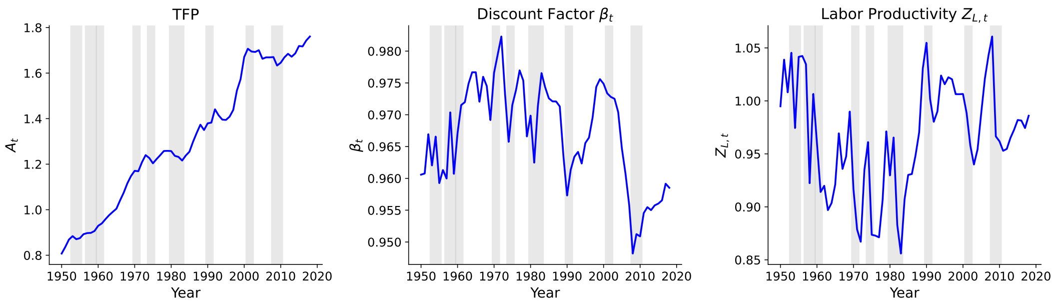

Figure 2: Recovered structural series from the data-anchored inversion

Notes: Top: aggregate TFP. Middle: saving-margin wedge β_t. Bottom: labor-efficiency wedge $Z_{L,t}$. The inversion uses observed output, capital, and hours. Shaded areas indicate NBER recession dates.

a secular decline from 1950 to 2019. The labor-efficiency wedge $Z_{L,t}$ fluctuates at business-cycle frequencies and captures movements in effective labor input beyond measured hours.

The MRCE-implied TFP tracks the representative-agent analogue closely at business cycle frequencies and correlates strongly with the Fernald utilization-adjusted series (Online Appendix E.1).

External labor discipline The second check asks whether the recovered labor-efficiency wedge is related to external labor-market margins not used in the inversion. At business-cycle frequencies, $Z_{L,t}$ is strongly correlated with the employment-population ratio and labor-force participation, the extensive margins it is designed to capture. The BLS labor composition index, by contrast, has the opposite cyclical sign: in recessions, low-productivity workers are disproportionately laid off, raising measured average quality, while $Z_{L,t}$ falls because utilization and matching deteriorate. The sign reversal supports the interpretation of Z_L as a utilization and matching margin rather than a demographic composition index.

Table 3: External validation: recovered $Z_{L,t}$ and labor-market indicators

External series	Level corr	HP-cycle corr
Employment-population ratio (EMRATIO)	0.48	0.81
Labor force participation rate (CIVPART)	0.28	0.54
Avg. weekly hours, manufacturing (AWHMAN)	0.42	0.41

Notes: The external series are not used in the inversion. HP-cycle correlations use annual data with $\lambda = 6.25$.

Saving-margin discipline The declining trend in β_t is not used as a target, so it needs external discipline. Figure 3 compares the recovered saving-margin wedge with three outside series: the personal saving rate, labor-force participation, and revolving consumer credit relative to GDP. The correlations are not a structural proof, but they help interpret β_t as a compact saving-side wedge that absorbs forces operating through the saving margin, including demographic shifts, financial liberalization, and precautionary motives.

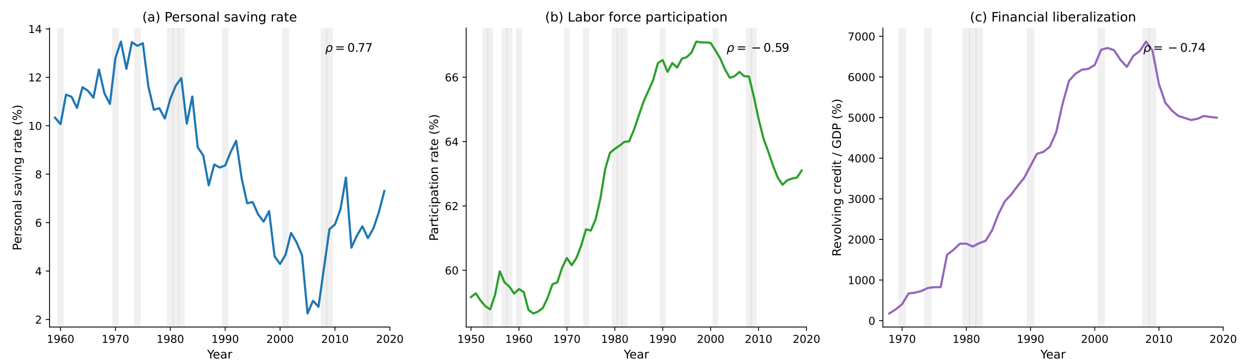

Figure 3: External discipline for the saving-margin wedge

Notes: External indicators from FRED, not used in the inversion. Panels show the U.S. personal saving rate, civilian labor-force participation, and revolving consumer credit as a share of GDP against the recovered β_t. Shaded areas indicate NBER recession dates.

A two-dimensional business cycle Figure 4 displays pairwise scatter plots of the recovered shocks. The most striking finding is the near-orthogonality of TFP with both $Z_{L,t}$ and β_t: technology evolves independently of the labor-saving block. The key cyclical correlation is between labor efficiency and patience, $\rho(\Delta \log Z_{L,t}, \Delta\beta_t) = -0.858$. Labor-market deterioration is systematically accompanied by higher saving, consistent with a precautionary channel. Together with the Krusell–Smith validation, the scatter plot shows that the inversion is not merely relabeling one residual: trend productivity and the labor-saving block occupy different dimensions of the data.

Selection robustness of the patience–productivity link. A natural concern is that the negative correlation $\rho(\Delta \log Z_{L,t}, \Delta\beta_t) = -0.858$ could reflect cyclical labor-force composition rather than a structural patience–productivity link: in expansions, lower-productivity workers may enter the labor force and dilute the measured average, while β_t moves separately, producing a spurious co-movement. We test this directly by regressing $\Delta\beta_t$ on $\Delta \log Z_{L,t}$ and adding three composition / participation controls in sequence: the civilian labor-force participation rate, the employment-population ratio, and the BLS nonfarm-business labor composition index. None of these series enters the inversion. Across 1951–2019, the partial coefficient on $\Delta \log Z_{L,t}$ moves from -0.092 ($t = -13.7$) in the bivariate specification to -0.155 ($t = -32.2$) once all three controls are included; the residualized correlation rises from

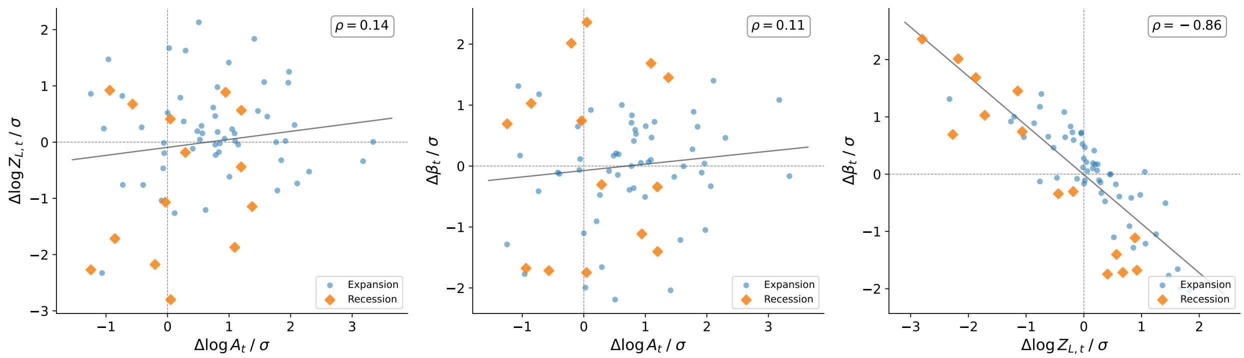

Figure 4: Pairwise scatter of recovered shock changes: the two-dimensional business cycle

Notes: Each point is one year. Blue circles are expansion years; orange diamonds are recession years. Changes are normalized by their unconditional standard deviations. The right panel shows the labor-saving tandem; the left and center panels show that TFP innovations are nearly orthogonal to the other two shocks.

-0.858 to -0.895, and an HP-cycle correlation on the levels delivers -0.850. The structural relationship therefore *sharpens* once cyclical selection is netted out, rather than collapsing: composition effects were partly attenuating the underlying patience–productivity link, not creating it. The cyclical co-movement of patience and labor productivity is a structural feature of the inversion, not a labor-supply artifact.

Cross-survey validation of the household inversion. We apply (32) to PSID 2005–2019 (8 waves, ∼4,000 households per wave) and (33) to SCF 1992–2019 (10 waves, 11,000–18,000 households per wave; the 1989 wave is excluded as a methodology outlier, see Table 4 notes). For SCF we clip β at $[0.5, 1.5]$ to bound the formula's tails when the implied leisure quantity $\kappa + \tilde{\xi}_t a_i - n_i$ flips sign. Table 4 reports population-weighted moments by wave for both surveys.

Table 4: Recovered structural parameters: PSID and SCF, sample-mean across waves

Source	Years	N/wave	$\bar{\beta}_{\text{cap}}$	$\bar{\beta}_{\text{med}}$	$\rho(\beta, z)$	$\rho(\beta, \log a)$	$\rho(z, \log a)$
PSID	2005–2019	4,091	0.894	1.015	−0.05	−0.32	+0.39
SCF	1992–2019	14,969	0.877	0.992	−0.28	−0.59	+0.22
Macro β (§4)	1950–2019	—	0.967	—	—	—	—

Notes: $\bar{\beta}_{\text{cap}} = E_w[\beta_i a_i R_i]/E_w[a_i R_i]$ averaged across waves. $\bar{\beta}_{\text{med}}$ is the wave-pooled median. Correlations are sample-mean across waves; standard deviations across waves are below 0.04 for $\rho(\beta, z)$ and below 0.08 for the gradients. Macro β is the 1950–2019 mean of AnalMacro1's macro inversion (`task2e_ABZL_inversion.csv`); the PSID 2005–2019 sub-window mean is 0.956 and the SCF 1992–2019 sub-window mean is 0.962. SCF 1989 is excluded: it is a methodology outlier (n=1,067 vs. 11,000–18,000 in subsequent waves; $\rho(\beta_i, \log a_i)$ sign-flips to +0.25 versus −0.51 to −0.63 in all later waves) due to first-survey weighting that the AHEAD redesign fixed in 1992.

Two surveys with disjoint samples and orthogonal identification strategies yield $\hat{\beta}_{\text{cap}}$ within 1.7 pp of each other (PSID 0.894 vs. SCF 0.877) and signed-equal cross-sectional gradients. Both are 7–9 pp below the macro $\beta_t = 0.967$ (1950–2019 mean; the PSID 2005–2019 sub-window has $\beta_t = 0.956$ and the SCF 1992–2019 sub-window has $\beta_t = 0.962$, so the magnitude of the gap is largely insensitive to the reference window). We unpack the time-series structure of these moments in Section 5.5 and discuss the implications in Section 5.7.

5 Macro and micro implications

With the structural shocks $(A_t, \beta_t, Z_{L,t})$ and household parameters (z_i, e_i, β_i) in hand, this section traces their economic content. We move from aggregate dynamics through the sorting-accelerator mechanism, then to conditional forecasting and tax counterfactuals, and finally to household-level evidence that the structural identification underlying those counterfactuals is invariant to the very tax reforms it studies.

5.1 Aggregate dynamics and the structural recession classification

The analytical aggregate equations imply three quantitative lessons. First, representative-agent aggregation is an excellent approximation in the baseline calibration. Investment decomposes into a representative-agent channel, a return-sorting channel, and a productivity-sorting channel; the representative-agent channel accounts for 93.2 percent of mean investment. This explains why the Krusell–Smith approximation works so well even though the exact state contains additional covariances.

Second, the omitted covariances still create a state-dependent amplification channel, and the analytical structure makes the generalized impulse response especially transparent. In a conventional heterogeneous-agent model a generalized impulse response must be computed by Monte Carlo, averaging simulated paths with and without the shock over draws of the idiosyncratic and aggregate history, precisely because the response depends on the entire distribution at the date of the shock (Koop et al., 1996). Here the response is governed by the three sufficient moments $\mathcal{M}$: two economies with the same $\mathcal{M}$ have identical impulse responses regardless of the rest of the distribution, so the state dependence is summarized by a three-dimensional object rather than an infinite-dimensional one. We compute generalized impulse responses to a 2 percent negative TFP shock from the 1952 distribution, when wealth inequality is low, and from the 2019 distribution, when wealth inequality is high. The impact effect is 3 percent larger from the 2019 state (-2.29% versus -2.22%), and the cumulative 10-year output loss is modestly larger (-20.4% versus -20.2%). The magnitude is small at the conservative baseline calibration, but the sufficient statistic is transparent: amplification rises with return-wealth sorting, summarized entirely by $\mathrm{cov}(e, a)$.

Third, the direct asset-pricing effect of return-wealth sorting is negligible at $\sigma_e = 0.01$. The interest-rate effect is approximately -0.001 basis points. This is a useful negative result. In the baseline calibration, return heterogeneity matters more as an amplification state variable than as a quantitatively important risk-premium channel. An Epstein–Zin extension, reported in the Online Appendix, is a more promising route for asset-pricing applications.

Structural recession classification. The exact inversion classifies recessions structurally—by identifying which component of the traced primitive path moves most—without narrative identification or external instruments. Table 5 reports recovered shocks around the seven major postwar NBER recessions falling within our 1950–2019 sample (1953–54, 1957–58, 1973–75, 1980, 1981–82, 1990–91, 2007–09), in standard-deviation units; three milder NBER recessions over the same sample (1960–61, 1969–70, 2001) produce negligible output declines and are omitted from the table. The classification rule assigns the label of the largest shock in $|\sigma|$ units; combined labels appear when multiple shocks exceed 1σ. The resulting typology is richer than supply-versus-demand: the 1957–58 and 1981–82 episodes combine labor and TFP channels, the 1973–75 episode features a large patience decline alongside modest TFP and labor movements, and the 2007–09 contraction is dominated by TFP. This heterogeneity in recession mechanisms illustrates the value of a three-shock framework.

Table 5: Structural recession classification from the exact inversion

Episode	ΔY (σ)	$\Delta \log A$ (σ)	$\Delta \log Z_L$ (σ)	$\Delta\beta$ (σ)	Classification
1953–54 (Korean War)	−1.12	−0.86	−1.72	−0.09	Labor-driven
1957–58 (Eisenhower)	−1.16	+0.34	−2.98	+2.05	Labor-driven
1973–75 (Oil Crisis)	−1.27	−1.63	−2.33	−2.44	TFP/Labor
1980 (Volcker I)	−0.68	−0.04	−1.07	+0.74	Labor-driven
1981–82 (Volcker II)	−1.33	−1.14	−2.18	+0.34	TFP/Labor
1990–91 (S&L Crisis)	−0.69	+1.31	−1.26	−0.48	Labor-driven
2007–09 (Great Recession)	−2.07	−1.24	−2.27	−2.16	TFP/Labor

Notes: All entries are peak-to-trough changes in units of each variable's unconditional standard deviation. Classification: "TFP-driven" when $|A|$ dominates; "Labor-driven" when $|Z_L|$ dominates; "Demand-driven" when $|\beta|$ dominates.

5.2 The sorting accelerator and cross-sectional dynamics

TFP growth and inequality The central distributional mechanism is a sorting accelerator. TFP growth raises returns, persistent return heterogeneity causes high-return households to become increasingly wealthy, and wealth concentration then raises the importance of return-wealth sorting in future dynamics. The relevant covariance, $\mathrm{cov}(e, a)$, is therefore highly persistent: the dominant eigenvalue governing distributional dynamics exceeds one in 100 percent of sample years, with mean 1.02.

Four distributional counterfactuals isolate the mechanism. The baseline simulation uses the historical paths of all recovered shocks. The TFP-only counterfactual keeps only the TFP path. The fixed-A counterfactual removes TFP growth. The all-fixed counterfactual holds all aggregate shocks fixed. Figure 5 reports the results. Both sorting channels grow substantially along the baseline path, but for different reasons. Return-wealth sorting grows through near-unit-root persistence, while productivity-wealth sorting grows through recession-driven bursts in idiosyncratic dispersion. Without TFP growth, return-wealth sorting stagnates; without countercyclical dispersion, productivity sorting loses its recession spikes.

Analytical decomposition The MRCE makes the sorting accelerator precise. In the return-sorting channel, persistence is governed by $\phi_e \rho_e (1 + \tilde{r})$, which is close to $1/\beta$ because return shocks are highly persistent. In the productivity-sorting channel, persistence is lower, but countercyclical dispersion raises the innovation rate during downturns. Figure 6 decomposes these forces. The secular component comes from persistent return heterogeneity; the cyclical component comes from recession-driven increases in idiosyncratic dispersion.

Who gains from TFP growth? Table 6 reports the impact of 1 percent TFP growth by wealth percentile. The top decile captures 81.6 percent of the aggregate wealth gain, exceeding its 77.2 percent wealth share, because the return-sorting and productivity-sorting channels compound. The top 1 percent alone captures 68.1 percent. The table is not a tax counterfactual; it is a sufficient-statistic calculation showing how the analytical channels allocate a common productivity improvement across the wealth distribution.

Historical gain shares The steady-state calculation is conservative because it holds the distribution fixed. Along the historical path, the distribution itself changes. Table 7 reports decade averages of TFP-driven wealth gains by wealth decile, and Figure 7 visualizes the evolving distribution. The top decile captures approximately 67.3 percent of TFP-driven wealth gains in the 1950s, rising to 69.4 percent by the 2010s. This increase reflects the endogenous accumulation of return-wealth sorting. As the wealth distribution concentrates

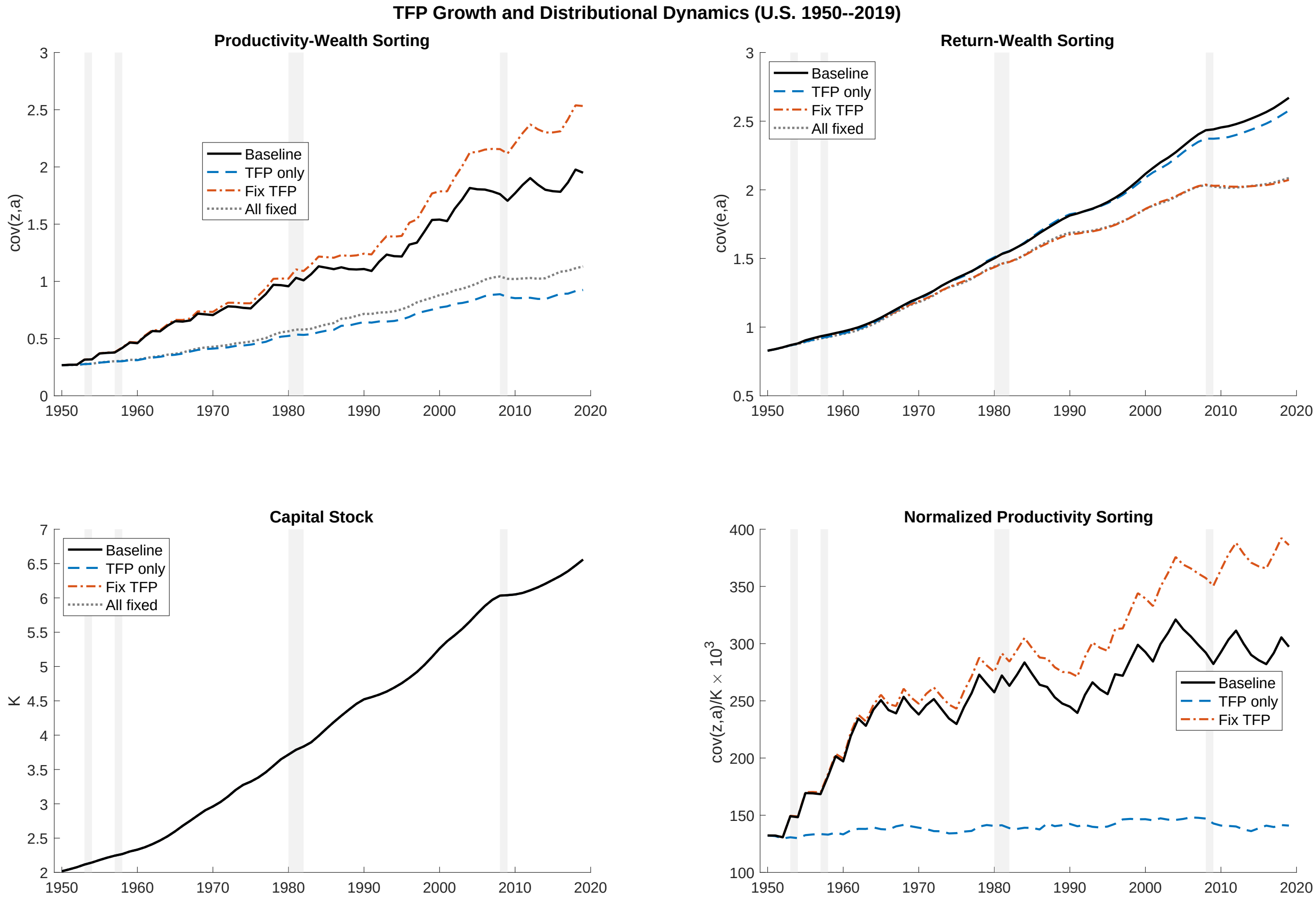

Figure 5: TFP growth and distributional dynamics

Notes: Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. Shaded areas indicate NBER recession dates.

through the Kesten mechanism, the return-sorting and productivity-sorting channels compound through the exact saving rule $a' = \beta R_i a + T$, and the top decile's share of any marginal productivity improvement grows accordingly.

The cumulative effect of the sorting accelerator is most visible in counterfactuals involving persistent shocks. The next subsection uses two historical capital-tax reforms to show that the near-unit-root sorting dynamics prevent mean reversion after a policy change, so distributional gaps opened by even transient reforms compound for decades.

Cross-sectional dynamics and consumption. The same historical simulation generates cross-sectional moments. Table 8 compares the model's wealth Gini with SCF-based estimates at survey years. The model captures the broad level of U.S. wealth inequality through the Kesten-process rule $a' = \beta R_i a + T$, with persistent return shocks and positive

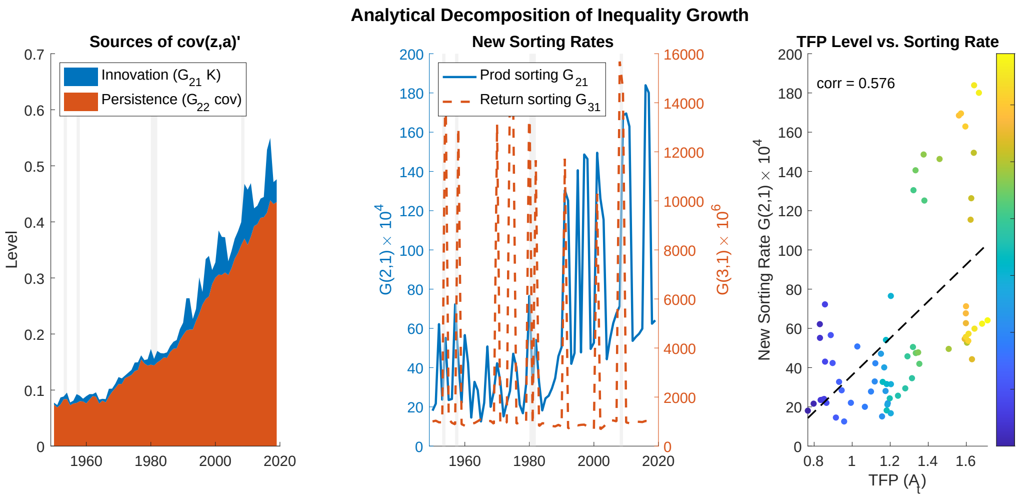

Figure 6: Analytical decomposition of distributional sorting

Notes: Left: innovation and persistence contributions to $\text{cov}(z, a)'$. Center: sorting-rate time paths for productivity and return shocks. Right: TFP level against productivity sorting. Shaded areas indicate NBER recession dates.

Table 6: Distributional impact of 1% TFP growth at steady state

Group	Wealth Share (%)	Gain Share (%)	Mean Gain (%)	Channel (%) Capital	Labor	Transfer
Bottom 10%	1.2	0.8	0.0806	65.1	18.9	16.0
10–20%	1.4	1.0	0.0850	67.9	19.8	12.3
20–30%	1.7	1.2	0.0873	69.4	20.2	10.4
30–40%	1.9	1.4	0.0893	70.6	20.5	8.9
40–50%	2.2	1.7	0.0913	71.7	20.6	7.6
50–60%	2.5	2.0	0.0932	72.8	20.7	6.5
60–70%	3.0	2.4	0.0954	74.0	20.7	5.3
70–80%	3.7	3.1	0.0982	75.4	20.5	4.1
80–90%	5.3	4.6	0.1019	77.2	20.1	2.8
Top 10%	77.2	81.6	0.1102	84.0	15.9	0.2
Top 5%	72.8	77.6	0.1138	84.2	15.7	0.1
Top 1%	63.1	68.1	0.1195	84.6	15.4	0.0

Notes: Gains computed from 1% TFP growth at the stationary distribution of simulated households. "Capital" is the gain from higher after-tax capital income; "Labor" is the gain from higher labor income; "Transfer" is the common transfer component.

transfers. It rises more smoothly than the data. This is a useful miss: time-varying return dispersion, top income growth, and housing-market divergence are natural candidates for the remaining trend.

Figure 8 summarizes the time-series path of wealth, consumption, and income inequality.

Table 7: Share of TFP-driven wealth gains by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.3	1.7	2.1	2.4	2.9	3.5	4.4	5.7	8.8	67.3	59.7	44.4
1960s	1.3	1.7	2.1	2.5	2.9	3.5	4.4	5.8	8.8	67.1	59.4	44.1
1970s	1.2	1.7	2.0	2.4	2.9	3.5	4.4	5.8	8.9	67.1	59.4	44.0
1980s	1.2	1.6	2.0	2.4	2.9	3.5	4.4	5.8	8.9	67.3	59.5	43.8
1990s	1.1	1.6	1.9	2.3	2.8	3.4	4.3	5.8	8.9	67.8	59.9	44.0
2000s	1.0	1.5	1.8	2.2	2.7	3.3	4.3	5.8	9.0	68.4	60.3	43.8
2010s	0.9	1.3	1.7	2.1	2.6	3.2	4.1	5.7	9.0	69.4	61.2	44.3
Full sample	1.1	1.6	1.9	2.3	2.8	3.4	4.3	5.8	8.9	67.8	59.9	44.0

Notes: Each entry reports the average share of aggregate TFP-driven wealth gains captured by the indicated wealth decile. Gains are computed period by period along the historical simulation.

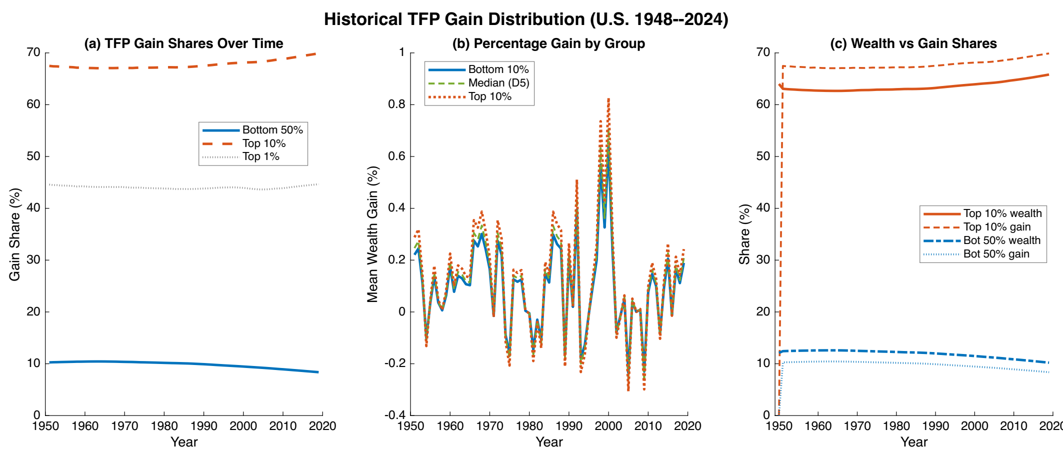

Figure 7: Historical distribution of TFP-driven gains

Notes: Period-by-period decomposition of the share of TFP-driven wealth gains by wealth percentile, U.S. 1950–2019.

Three patterns stand out. First, wealth concentration rises monotonically from 0.827 to 0.876. The driver is the near-unit-root persistence of return-wealth sorting, the same Kesten mechanism that generates the sorting accelerator. Second, consumption inequality also rises (from 0.281 to 0.356). The consumption rule $c_i \propto (1-\beta)R_i a_i$ loads directly on wealth, so cross-sectional dispersion is amplified as the wealth distribution concentrates. Third, income inequality is flatter (mean Gini 0.404). Two offsetting forces are at work: rising wealth inequality pushes capital income apart, but the labor component, which scales with $Z_{L,t}z_i$ rather than wealth, compresses during recessions when low-productivity workers exit. The divergence between consumption and income Gini is a testable implication of the MRCE's exact policy structure.

The time-series Gini compresses the distribution into one number. Figure 9 overlays the

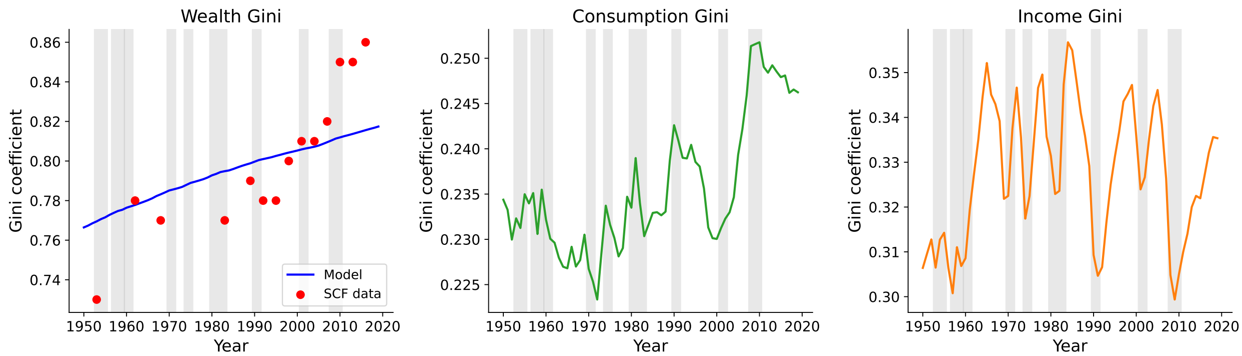

Figure 8: Historical cross-sectional distributional dynamics

Notes: Historical simulation using the recovered aggregate shocks and exact household policies. The wealth panel includes SCF-based estimates as external benchmarks. Shaded areas indicate NBER recession dates.

Table 8: Wealth Gini: model vs. data

Year	Data	Model	Year	Data	Model
1953	0.73	0.830	1998	0.80	0.864
1962	0.78	0.838	2001	0.81	0.865
1968	0.77	0.843	2004	0.81	0.867
1983	0.77	0.855	2007	0.82	0.869
1989	0.79	0.858	2010	0.85	0.871
1992	0.78	0.860	2013	0.85	0.873
1995	0.78	0.862	2016	0.86	0.875

Notes: Data are SCF-based estimates reported by Kuhn and Ríos-Rull (2025). Model moments are from the historical simulation using exact household policies.

full cross-sectional wealth distribution (normalized by mean wealth) in 1980 and 2015. The 2015 distribution exhibits a fatter right tail and a thinner middle: the mass between 0.5 and 1.5 times mean wealth declines while the upper tail extends. This is the distributional signature of the sorting accelerator: three decades of persistent return heterogeneity stretch the Pareto tail while the additive transfer T keeps the lower tail anchored.

Table 9 reports consumption and income moments in normal times and across recession episodes. Two patterns emerge. First, consumption inequality rises in recessions (+3.49% in CV on average) while income inequality falls (−5.42%). The consumption response reflects the model's exact policy structure: when β_t falls (rising impatience), the consumption share $(1-\beta_t)/(1+\theta)$ increases, amplifying the wealth-driven component $R_i a_i$ in cross-sectional consumption. Income compression reflects declining $Z_{L,t}$, which reduces labor-income dispersion by scaling down the $Z_L z$ term. Second, the magnitude of the consumption response varies sharply across episodes. The Great Recession stands out: consumption CV rises 26.5% above normal times, driven by the interaction of high wealth concentration ($\text{cov}(e, a)$ at its

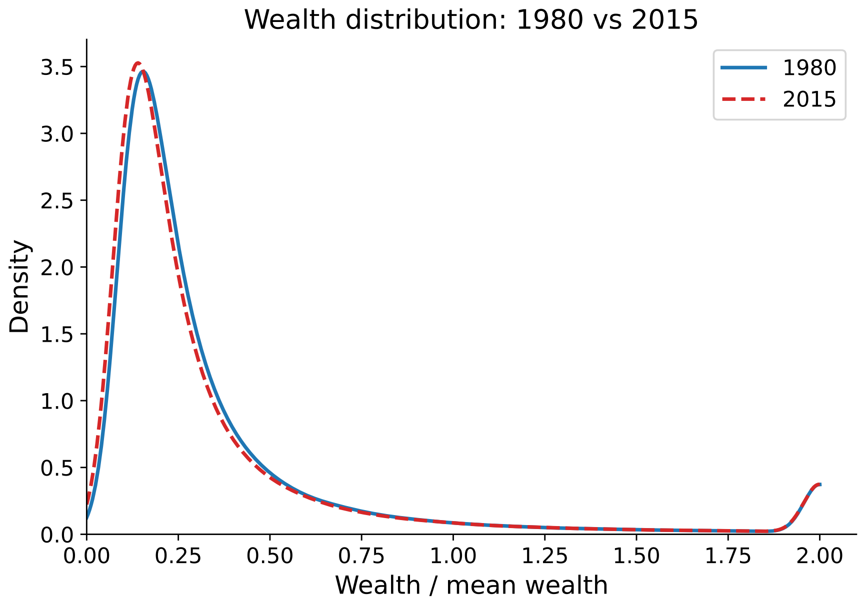

Figure 9: Normalized wealth distribution: 1980 vs. 2015

Notes: Kernel density estimates of wealth divided by mean wealth from the historical household simulation. Values are winsorized at twice mean wealth for readability.

sample maximum) with a sharp fall in the saving-margin wedge. By contrast, the 1953–58 recessions show consumption CV *below* normal, reflecting the lower wealth concentration of the early postwar period.

Table 9: Cross-sectional distributional moments: normal times vs. recessions

	Consumption				Income			
	CV	Gini	Skew	Kurt	CV	Gini	Skew	Kurt
Normal	1.009	0.334	8.19	92.3	1.798	0.408	11.75	178.7
1953–54 (Korean War)	1.003	0.291	10.28	134.2	1.738	0.359	12.89	203.6
1957–58 (Eisenhower)	0.973	0.303	9.42	118.1	1.600	0.363	12.30	189.7
1973–75 (Oil Crisis)	0.897	0.331	7.13	75.9	1.708	0.402	11.94	185.0
1980 (Volcker I)	0.928	0.318	8.15	92.1	1.742	0.394	11.67	167.9
1981–82 (Volcker II)	0.967	0.327	8.16	92.1	1.671	0.393	11.84	182.8
1990–91 (S&L Crisis)	1.168	0.340	9.58	114.8	1.696	0.392	11.99	187.3
2007–09 (Great Recession)	1.276	0.365	9.34	107.4	1.744	0.411	11.80	191.5

Notes: Cross-sectional moments from simulating households along the historical equilibrium path using the three-shock inversion with countercyclical dispersion. CV = coefficient of variation. Normal times are non-NBER-recession years; recession moments are averaged within each episode.

Consumption dynamics The MRCE also yields a closed-form aggregate consumption equation,

$$C_t = \frac{1-\beta_t}{1+\theta}\left[(1+\tilde{r}_t+\tilde{w}_t Z_{L,t}\tilde{\xi}_t)K_t + \tilde{r}_t \operatorname{cov}_t(e,a) + \tilde{w}_t Z_{L,t}\tilde{\xi}_t \operatorname{cov}_t(z,a)\right] + \kappa\tilde{w}_t Z_{L,t},$$

where $\tilde{\xi}_t \equiv \xi/A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient (see Online Appendix B). The individual MPC out of a one-time transfer is $(1-\beta_t)/(1+\theta)$, independent of wealth; it is a constant that varies only with the aggregate saving-margin wedge. Figure 10 reports three objects. The left panel shows the consumption-output ratio, which rises from 0.69 in the 1980s to 0.78 by 2010, tracking the secular decline in β_t (lower patience $\rightarrow$ higher consumption share). The center panel plots the annual MPC, which mirrors β_t: the MPC spikes during the 2007–09 recession when patience falls sharply. The right panel shows the cumulative MPC at steady state reaching approximately 25% after 20 years. The low annual MPC ($\sim 0.665\%$) is a scope condition: the MRCE isolates aggregation and sorting rather than liquidity-channel consumption responses (Kaplan et al., 2018).

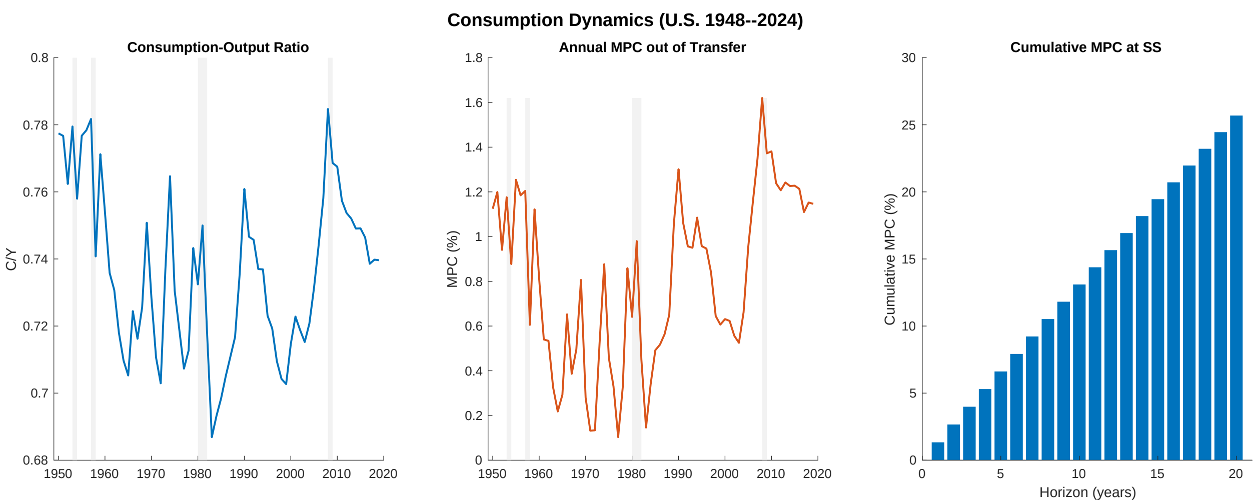

Figure 10: Consumption dynamics and transfer MPC

Notes: Left: consumption-output ratio. Center: annual MPC out of a one-time transfer implied by $(1-\beta_t)/(1+\theta)$. Right: cumulative steady-state MPC by horizon.

5.3 Conditional forecasting

A distinctive advantage of the MRCE is that conditional forecasting is computationally direct. In conventional heterogeneous-agent models, forecasting requires re-solving the household dynamic program or estimating a perceived aggregate law of motion, because policy functions depend on beliefs about future aggregate states. In the MRCE, household policies are backward-looking and linear in wealth. Given any future sequence for $(A_t, \beta_t, Z_{L,t})$, the model produces output, capital, labor, prices, consumption, investment, and the distributional state by forward simulation alone.

Rolling forecasts We exploit this property with a rolling forecast exercise. For each origin year $t \in \{1965, \ldots, 2009\}$, the inversion recovers structural shocks from data available up to t, and the model is simulated forward holding shocks at their last recovered values. The resulting forecast jointly produces output, capital, labor, prices, consumption, investment, and the distributional state, all satisfying the production function, capital accumulation, and market clearing simultaneously.

Figure 11 compares the MRCE forecast to a recursive AR benchmark that forecasts the structural inputs using the same information set and aggregates them through the same production function. At both one- and three-year horizons, the MRCE has lower RMSE for output (3.1% vs. 7.0% at $h = 1$; 5.1% vs. 9.4% at $h = 3$) and labor (2.6% vs. 4.2%; 5.4% vs. 6.1%). For capital, the AR benchmark exploits the high persistence of the raw capital stock ($\rho = 0.99$) and is difficult to beat at short horizons. The MRCE's advantage is structural: it produces a jointly consistent forecast of all prices and quantities, while the AR inputs evolve independently and are not required to satisfy any equilibrium condition.

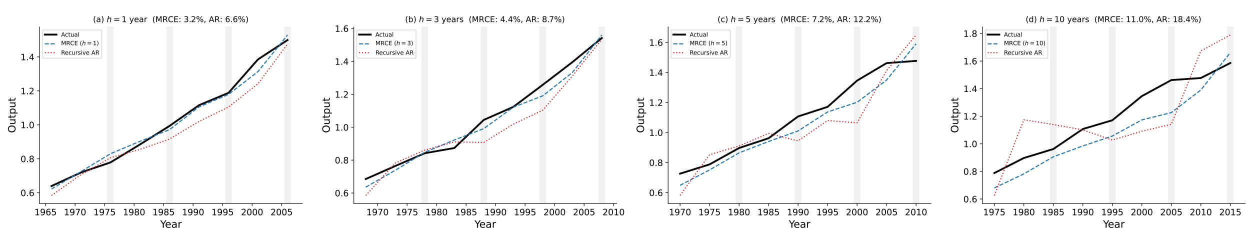

Figure 11: Rolling forecast errors by horizon

Notes: For each origin year, the model is simulated forward holding recovered shocks at their last values. The recursive AR benchmark forecasts structural inputs using the same information set and aggregates through the same production function. Panel titles report RMSEs. Shaded areas indicate NBER recession dates.

5.4 Distributional impact of tax reforms

The near-unit-root persistence of wealth sorting implies that even temporary changes in capital taxation can produce almost permanent shifts in the wealth distribution. We test this prediction using four historically abrupt capital tax changes from both directions of the political spectrum: two cuts (Reagan ERTA 1981, τ_K from 0.380 to 0.375; Bush EGTRRA 2001, τ_K from 0.395 to 0.355) and two increases (OBRA 1993 under Clinton, τ_K from 0.340

to 0.360; ATRA 2013 under Obama, τ_K from 0.278 to 0.310).[3]

For each episode, two full agent-based simulations (500,000 agents) use identical aggregate shocks and idiosyncratic histories. The counterfactual is *reform-only*: $\tau^{\text{cf}}_{K,t}$ equals the factual path minus the named reform's reform-year shock from the reform date onward, leaving all other reforms intact. For example, the OBRA-only counterfactual subtracts $+0.020$ from $\tau_{K,t}$ for $t \geq 1993$ but keeps the Bush 2001 cut, ATRA 2013 increase, and TCJA 2017 cut in both factual and counterfactual paths. This isolates the marginal effect of each named reform rather than the cumulative effect of "no policy change after the reform date." Figures 12 and 13 report the cuts and increases respectively, in the same panel layout.

Several features of Figure 12 merit attention. First, the Gini gap opens immediately at each reform date and continues to widen for decades, confirming the near-unit-root prediction: once the wealth distribution is pushed toward greater concentration, persistence compounds the initial shift rather than allowing mean reversion. Second, the wage panel (row 4) reveals the growth side of the tradeoff: lower capital taxes stimulate capital accumulation, raising the marginal product of labor and hence equilibrium wages, so all households benefit from higher wages even as inequality rises. Third, the return-wealth covariance (row 5) diverges at each reform date, tracing the sorting mechanism directly: lower τ_K raises after-tax capital returns, which disproportionately benefit high-wealth households, accelerating $\text{cov}(e, a)$ growth.

Figure 13 confirms that the same mechanism runs in reverse for tax increases. After OBRA 1993 raises τ_K by $+0.020$ (from 0.340 to 0.360), the factual wealth Gini falls relative to the reform-only counterfactual, the return-wealth covariance compresses, and output and wages contract. The Bush (-0.040) and ATRA ($+0.032$) reform-year shocks dominate the Reagan (-0.005) and OBRA ($+0.020$) shocks in magnitude, but cumulative distributional effects are governed jointly by shock size and post-reform time horizon: OBRA 1993 has 26 post-reform years in the sample, ATRA 2013 only 6. The reform-only design isolates each reform from all subsequent policy: in particular, the OBRA panel is no longer contaminated by the post-1993 Bush 2001 and TCJA 2017 cuts.

Panel A of Table 10 quantifies the growth-versus-inequality tradeoff for the cuts under the reform-only counterfactual. Stacking both reforms, the combined isolated effect by 2019 is a 1.01% output gain and 0.87% wage increase, offset by a 0.20 percentage point wealth

[3] Annual effective tax rates ($\tau_{L,t}, \tau_{K,t}$) used in the inversion and counterfactuals are not directly observed at annual frequency. We construct them as a year-by-year stylized series consistent with the decade averages of McDaniel (2007) and the timing of major federal legislation (Korean War surtax, Kennedy 1964, ERTA 1981, TRA 1986, OBRA 1993, EGTRRA 2001, JGTRRA 2003, ARRA 2009, ATRA 2013, TCJA 2017). The full series is in `exact_ge_mapping/data/effective_tax_rates.csv`; see `build_tax_data.py` for the construction. Counterfactual conclusions about *differences* between factual and reform-only paths are robust to reasonable variation in the tax-rate series because both scenarios are subject to the same construction.

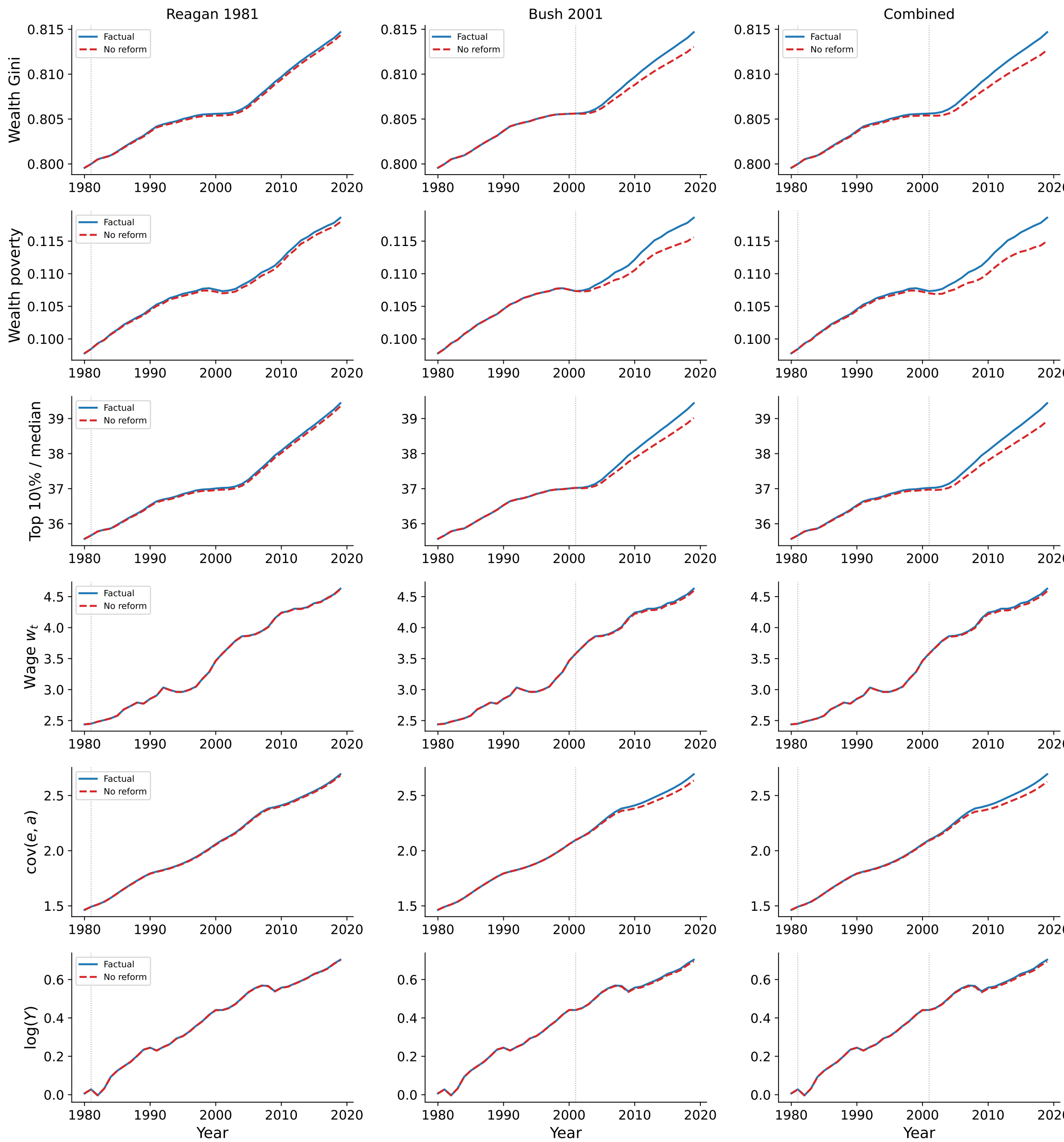

Figure 12: Macroeconomic impact of capital tax cuts (Reagan 1981, Bush 2001)

Notes: 500,000-agent simulation, 1980–2019. Columns: Reagan 1981, Bush 2001, and combined cuts. Rows show wealth Gini, wealth poverty (share of households with wealth below half the median), top 10%/median ratio, wage, return-wealth covariance, and output. Solid blue: factual; dashed red: counterfactual (no reform). Capital evolves endogenously; all GE prices re-solved each period.

Gini increase, a 1.30% rise in the top 10%/median ratio, and a 3.11% increase in the wealth-poverty rate. The income share of the top 1% rises by 0.40 percentage points while the bottom 50% loses 0.18 points. The magnitudes are smaller than under the freeze-pre-reform design (which conflates each reform with all subsequent policy), but the signs and rankings are the same: Bush EGTRRA delivers most of the cumulative effect because its reform-

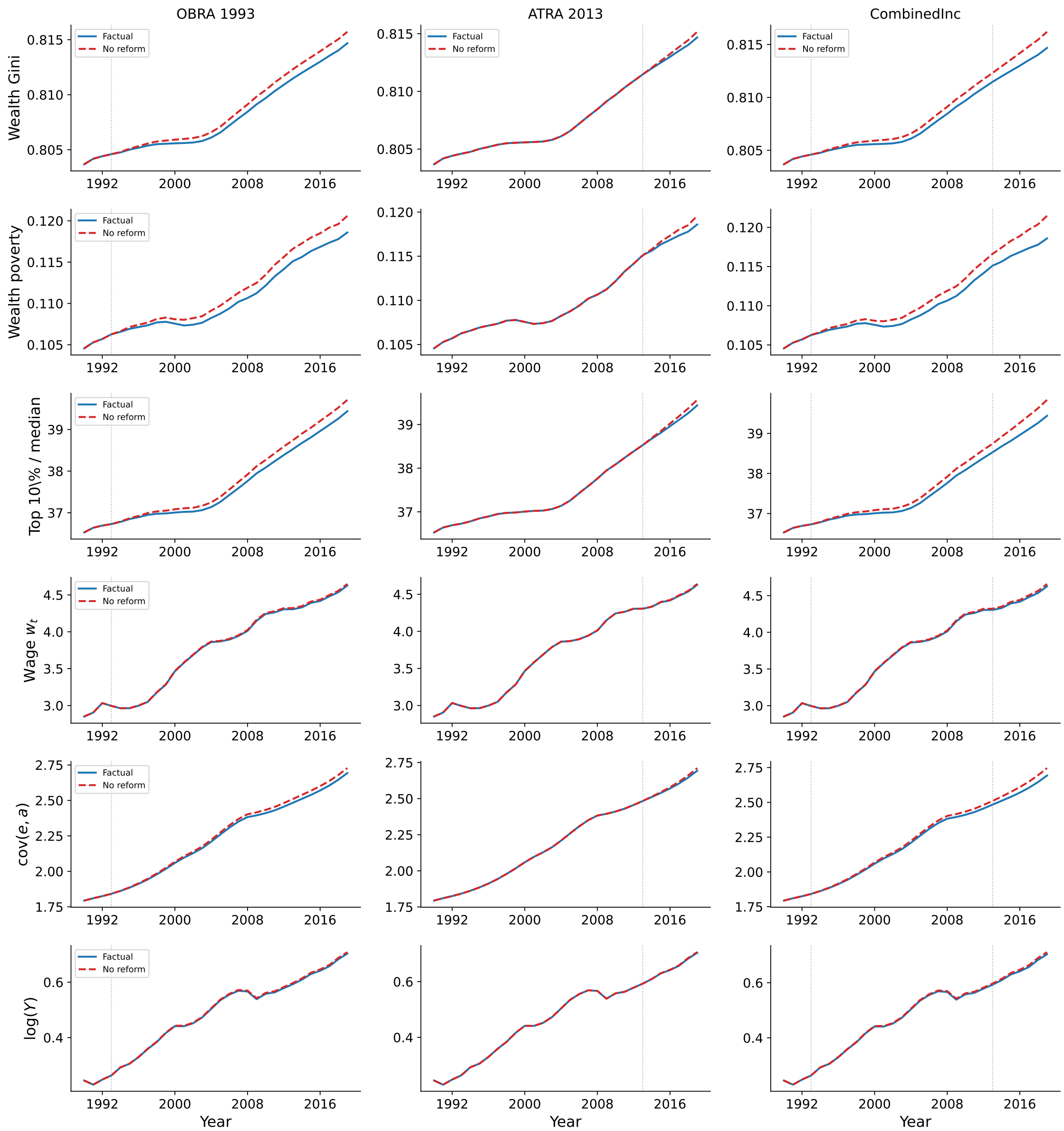

Figure 13: Macroeconomic impact of capital tax increases (OBRA 1993, ATRA 2013)

Notes: Same simulation and panel layout as Figure 12. Columns: OBRA 1993, ATRA 2013, and combined increases. Solid blue: factual; dashed red: reform-only counterfactual (subtract only the named reform's reform-year τ_K shock from the factual path going forward, keeping all subsequent reforms intact). The horizontal axis starts at 1990 to focus on the post-reform window.

year cut (-0.040) is much larger than Reagan ERTA's (-0.005). Even under the cleaner interpretation, the bottom decile gains in absolute income levels but loses share, as the sorting mechanism channels the growth dividend toward the top.

Panel B of Table 10 reports the symmetric numbers for the two tax-increase episodes under the same reform-only design. Because OBRA 1993 lies 17 years before 2010 and

ATRA 2013 lies 3 years *after* 2010, we use 2002 (9 years post-OBRA, before ATRA) and 2019 (terminal) as the horizons. Signs are uniformly negative for output, wage, and inequality measures, mirroring the cuts. By 2019 the combined isolated effect of both increases is -0.77% output, -0.67% wage, -0.15 pp wealth Gini, -1.00% top 10%/median, and -2.37% wealth poverty. The income share of the top 1% falls by 0.48 percentage points and the bottom 50% gains 0.21 points: the same sorting mechanism runs in reverse, redistributing toward the bottom when capital is taxed more heavily. OBRA 1993 carries most of the cumulative effect because it has 26 post-reform years in the sample versus only 6 for ATRA 2013.

The exercise establishes that the macro side's tax counterfactuals are not skewed toward one political direction: the same near-unit-root sorting mechanism produces decades-long distributional drift after both major cuts and major increases, with appropriately signed effects. Read together, the 1981, 1993, 2001, and 2013 episodes span the full postwar range of capital tax policy, and each leaves a structural footprint that the analytical framework traces directly. The magnitudes hold (z_i, e_i, β_i) fixed across the policy change; Section 5.7 discusses the scope of this maintained hypothesis.

5.5 Household parameter distributions and sorting at the cross-section

The macro tax counterfactuals just reported take (z_i, e_i, β_i) recovered from PSID and SCF as policy-invariant primitives. We now turn to those primitives directly: their lifecycle profiles, three-decade trends, variance composition, and age-conditional sorting. Section 5.6 then returns to the same data to test whether the structural parameters actually shift around the reforms studied in Section 5.4.

Lifecycle profiles. The cross-section reveals a clean lifecycle pattern in β that the macro aggregate masks. Figure 14 reports population-weighted means of (z_i, e_i, β_i) by 5-year age bin, pooling all waves within each survey.

Three patterns are robust across both surveys.

Productivity z_i traces a Mincer hump. Median z rises from $\sim$0.5 (SCF) or $\sim$0.9 (PSID) at age 27 to a peak near age 50, then declines. The level difference between surveys reflects different working-age conditioning (PSID requires positive wages and hours; SCF includes more part-time and transitioning workers).

Discount factor β_i declines with age. In PSID, the population-weighted mean β falls from 1.51 at age 27 to 1.21 at age 62; the median falls from 1.09 to 0.96. In SCF the decline is

Table 10: Growth vs. inequality: capital tax reforms (reform-only counterfactuals)

Panel A: Tax cuts	Reagan 1981		Bush 2001		Combined	
Δ	2010	2019	2010	2019	2010	2019
Growth (%)						
Output	+0.13	+0.16	+0.44	+0.84	+0.58	+1.01
Wage	+0.12	+0.14	+0.42	+0.73	+0.54	+0.87
Inequality measures						
Wealth Gini (pp)	+0.03	+0.03	+0.09	+0.16	+0.12	+0.20
Top 10%/median (%)	+0.18	+0.22	+0.56	+1.07	+0.74	+1.30
Wealth poverty (%)	+0.46	+0.55	+1.47	+2.59	+1.90	+3.11
Income share shift (pp)						
Top 1%	+0.04	+0.04	+0.41	+0.36	+0.45	+0.40
Top 10%	+0.04	+0.04	+0.38	+0.35	+0.42	+0.39
Bottom 50%	−0.02	−0.02	−0.17	−0.16	−0.18	−0.18
Bottom 10%	−0.00	−0.00	−0.02	−0.02	−0.02	−0.02
Panel B: Tax increases						
	OBRA 1993		ATRA 2013		Combined	
Δ	2002	2019	2002	2019	2002	2019
Growth (%)						
Output	−0.19	−0.51	0.00	−0.26	−0.19	−0.77
Wage	−0.15	−0.44	0.00	−0.24	−0.15	−0.67
Inequality measures						
Wealth Gini (pp)	−0.04	−0.10	0.00	−0.05	−0.04	−0.15
Top 10%/median (%)	−0.25	−0.68	0.00	−0.32	−0.25	−1.00
Wealth poverty (%)	−0.74	−1.67	0.00	−0.82	−0.74	−2.37
Income share shift (pp)						
Top 1%	−0.14	−0.16	0.00	−0.33	−0.14	−0.48
Top 10%	−0.14	−0.16	0.00	−0.31	−0.14	−0.47
Bottom 50%	+0.06	+0.07	0.00	+0.14	+0.06	+0.21
Bottom 10%	+0.01	+0.01	0.00	+0.02	+0.01	+0.03

Notes: Reform-only counterfactuals: each CF subtracts only the named reform's reform-year τ_K shock from the factual path going forward, leaving all other reforms intact. Reform-year shocks: Reagan −0.005, Bush −0.040, OBRA +0.020, ATRA +0.032. Growth and "Top 10%/median" / "Wealth poverty" rows: (factual − counterfactual) / counterfactual × 100. "Wealth Gini" row: percentage-point shift in Gini coefficient ×100. "Wealth poverty" is the fraction of households with wealth below half the median. Income share shifts in percentage points. "Combined" stacks the panel's two reform shocks additively. ATRA 2013 entries at 2002 are zero because the reform has not yet occurred.

smoother: the mean drops from 1.11 at age 27 to 0.95 at age 72, and the median from 1.07 to 0.95. Young households appear to "save more than the macro return predicts" ($\beta > 1$):

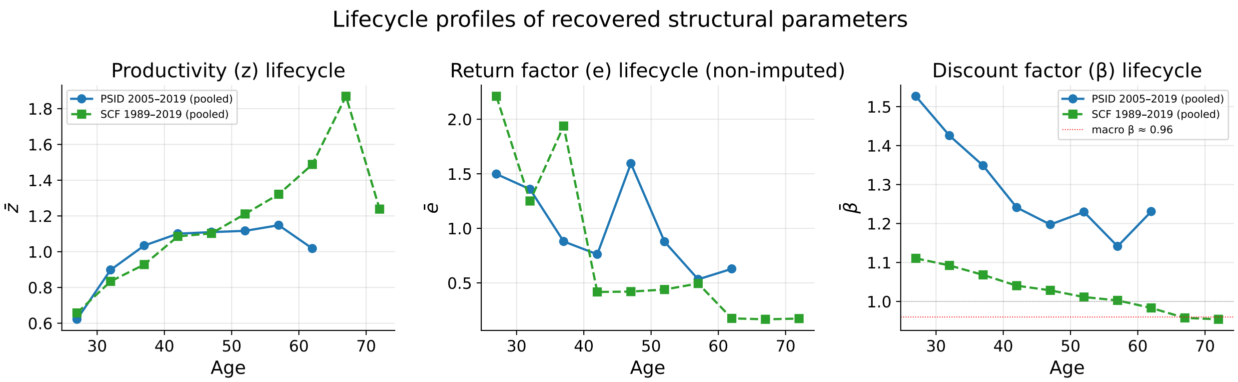

Figure 14: Lifecycle profiles of (z_i, e_i, β_i), pooled across waves

Notes: 5-year age bins from 25 to 74 (where sample size permits). PSID pools waves 2005–2019; SCF pools waves 1992–2019. Population-weighted means using survey weights. e panel uses non-imputed observations only.

they are accumulating wealth faster than the macro r alone would explain. This is consistent with housing appreciation, family transfers, and front-loaded retirement contributions, all of which the model's saving identity attributes to β_i. Late-life β_i converges to the macro level ~ 0.96 from above, consistent with mature households whose wealth dynamics are governed primarily by the model's gross return.

Wealth a_i rises monotonically with age in both surveys. Median wealth grows tenfold across the life cycle in PSID and 80-fold in SCF (whose right tail captures retirement-age accumulation that PSID misses).

Distributional dynamics over three decades. The 10 SCF waves and 8 PSID waves provide direct evidence on how the cross-section of structural parameters evolves over nearly three decades. Figure 15 plots four time series; panel (a) overlays the wave-by-wave $\hat{\beta}_{\text{cap,t}}$ in PSID and SCF against the macro β_t.

Heterogeneity in β widens over time. The cross-sectional spread $P_{90}(\beta_i) - P_{10}(\beta_i)$ in SCF rises from 0.27 in 1992 to 0.46 in 2019 (Figure 15b), a 70% increase concentrated in the post-2007 period (pre-2007 average ≈ 0.30; post-2007 average ≈ 0.43). PSID's spread rises from 1.55 in 2005 to 1.73 in 2019. The level difference between surveys reflects PSID's wider β range (PSID has unclipped saving-equation residuals; SCF clips the intratemporal-FOC tails); the trend is the same. Household-level saving behavior has become more disparate over the same three decades that wealth concentration rose, consistent with the macro paper's sorting accelerator.

The wealth gradient of β is large and stable. $\rho(\beta_i, \log a_i)$ stays in $[-0.51, -0.63]$ in SCF post-1992 and around -0.30 in PSID across all waves (Figure 15c). The wealthy persistently

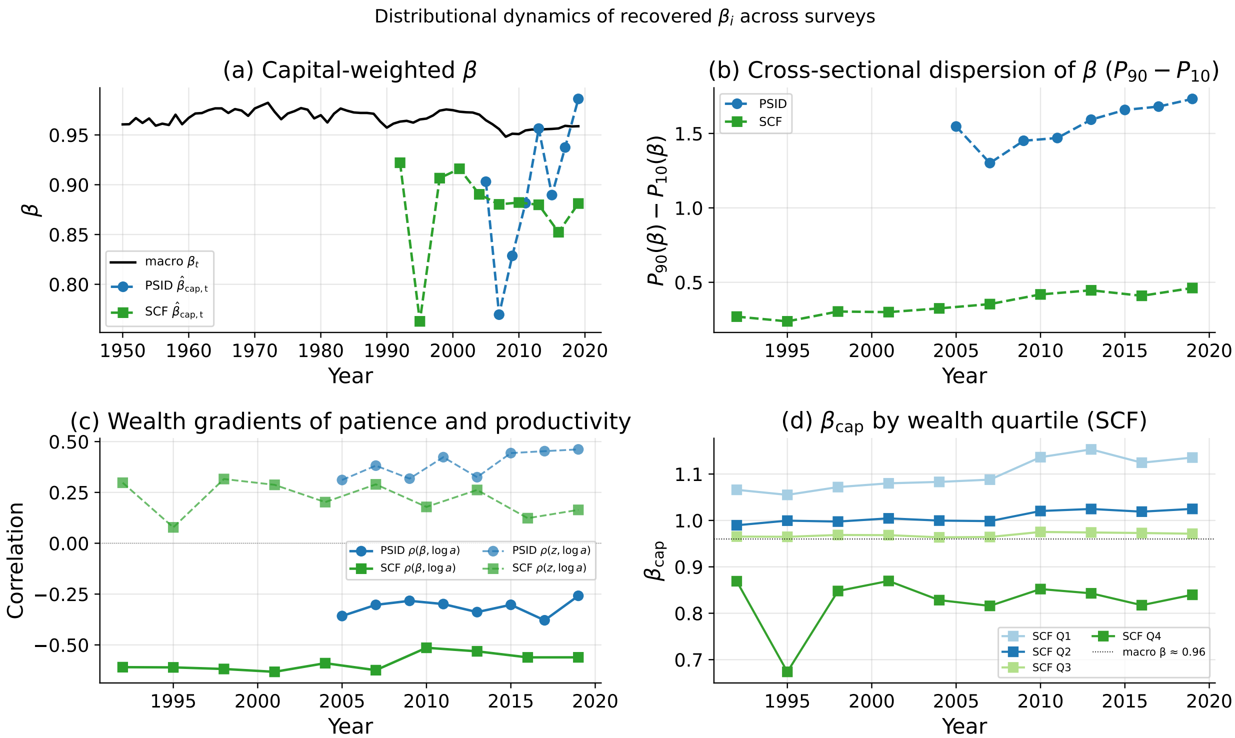

Figure 15: Distributional dynamics of recovered β_i over three decades

Notes: (a) Capital-weighted $\hat{\beta}_{\mathrm{cap,t}}$ in PSID (biennial 2005–2019) and SCF (triennial 1992–2019), with the macro β_t overlaid. (b) Cross-sectional dispersion $P_{90}(\beta_i) - P_{10}(\beta_i)$ within each wave. (c) Wealth gradients: $\rho(\beta_i, \log a_i)$ (solid) and $\rho(z_i, \log a_i)$ (dashed) within each wave. (d) Capital-weighted $\hat{\beta}_{\mathrm{cap}}$ by within-survey wealth quartile, SCF only.

dissave at $\beta < 0.86$ while the poor accumulate at $\beta > 1.10$. Table 11 reports the underlying quartile decomposition.

PSID Q4 wealth *shrinks* at -1.9%/yr, giving median $\beta = 0.93$ and capital-weighted $\beta_{\mathrm{cap}} = 0.86$; SCF Q4 has median $\beta = 0.92$ and $\beta_{\mathrm{cap}} = 0.82$. Because the capital-weighted aggregate is dominated by Q4 (which holds $\sim$80% of survey-aggregate wealth), the survey $\hat{\beta}_{\mathrm{cap}}$ inherits Q4's value of $\sim$0.86–0.88. The bottom three quartiles look conventional: rapid accumulation at the bottom, β falling toward the macro level as wealth rises. The top quartile drives the macro–micro gap.

Productivity-wealth sorting evolves differently across surveys. PSID's $\rho(z_i, \log a_i)$ rises from 0.31 in 2005 to 0.46 in 2019; SCF's declines from 0.30 in 1992 to 0.16 in 2019 (Figure 15c, dashed lines). The divergence is informative. PSID conditions on working-age households with valid wages: among them, productivity sorting strengthens over time, confirming the macro paper's sorting accelerator at the household level. SCF includes the very top of the wealth distribution that PSID misses; at that top, wealth is increasingly capital-gains and inheritance driven rather than wage driven, so the SCF z-a correlation weakens secularly.

Table 11: Discount factor by wealth quartile (PSID and SCF, all waves pooled)

Quartile	median a	median Δa %/yr	median β	β_{cap}
PSID 2005–2019 (biennial saving equation)				
Q1 (poorest)	0.20	+41.7	1.32	1.59
Q2	1.25	+9.3	1.05	1.19
Q3	3.99	+2.4	0.98	1.04
Q4 (top)	15.30	−1.9	0.93	0.86
SCF 1992–2019 (intratemporal FOC, cross-sectional)				
Q1 (poorest)	0.54	—	1.14	1.10
Q2	5.17	—	1.01	1.01
Q3	23.93	—	0.98	0.97
Q4 (top)	433.87	—	0.93	0.82

Notes: Quartiles formed within each survey using sample wealth a_t. Median a and median Δa are unweighted within quartile (the SCF dealer-sample weights tiny at the top and the PSID weights roughly uniform; weighted medians within Q4 mask its dissaving). Median a in model units (1 unit ≈ \$38,545, 2021 dollars). Δa is the median annualized geometric biennial wealth growth rate $(a_{i,t+2}/a_{i,t})^{1/2} - 1$ (PSID only). $\beta_{\text{cap}} = E_w[\beta_i a_i R_i]/E_w[a_i R_i]$ within quartile, weighted. Computed by `micro_diagnostics.py`.

The MRCE's second sorting covariance $\text{cov}(e, a)$ at the macro level absorbs this top-decile decoupling.

Variance decomposition: fixed effect, lifecycle, and noise. The PSID panel sharpens the question of where each parameter's variation comes from. Of the total cross-sectional variance, how much is a permanent individual fixed effect, how much is a within-person lifecycle trend, how much comoves with the wave (aggregate), and how much is idiosyncratic noise? Table 12 reports the decomposition.

Table 12: Variance decomposition of recovered parameters (PSID panel, 6,943 persons / 30,241 obs)

Parameter	Total var.	Person FE	Age (lifecycle)	Wave (agg.)	Idiosyncratic
$\log z$	0.678	76.9%	0.3%	0.3%	22.5%
β	3.297	29.6%	0.0%	0.1%	70.2%

Notes: Each row decomposes the total panel variance into a person fixed effect (variance of person-mean), a within-person age trend (variance explained by a third-order polynomial in age after demeaning within person), a wave/aggregate component (variance of within-person residual averaged by wave), and an idiosyncratic residual. Restricted to PSID persons observed in ≥ 2 waves; β is unclipped (the wide dispersion in PSID's biennial saving identification reflects wealth-shock noise). Computed by `micro_diagnostics.py` from the panel CSVs.

The verdicts contrast sharply. *For* $\log z$*, the fixed effect dominates.* Person FE absorbs

77% of variance; idiosyncratic shocks 22%; the within-person life-cycle trend only 0.3%. Productivity is overwhelmingly a permanent type, with within-person AR(1) of 0.77 biennial (0.88 annualized, close to the macro calibration $\rho_z = 0.95$). The MRCE's AR(1) z process matches the within-person idiosyncratic component well (its calibrated unconditional sd ~ 0.39 matches the within-person sd $\approx 0.83\sqrt{0.22} \approx 0.39$), but the data also reveal a much wider permanent component (sd $\approx 0.83\sqrt{0.77} \approx 0.73$) absent from the macro calibration. The cross-sectional Mincer hump in z by age (Figure 14) reflects partly within-person life-cycle accumulation but mostly between-person composition: different age groups draw from different fixed-effect distributions.

For β, idiosyncratic noise dominates. Person FE absorbs only 30% of variance; idiosyncratic shocks 70%; the within-person life-cycle trend essentially zero ($< 0.1\%$). Within-person biennial AR(1) is -0.12 (annualized -0.34): individuals do not have stable β paths, only mildly mean-reverting noise around their personal average. The lifecycle decline of β visible in the cross-sectional age profiles is therefore mostly a between-person composition effect, not within-person aging: at older ages the sample shifts toward households whose fixed-effect component happens to be lower, not toward individuals who became less patient. The fixed-effect component (the 30%) reflects cross-sectional preference heterogeneity; the dominant idiosyncratic component reflects realized wealth shocks (housing appreciation, business income volatility, family transfers) that the saving identity attributes to β and that mean-revert at the individual level.

The wave (aggregate) component is tiny for both. Less than 0.3% of $\log z$ or β variance comoves with the wave. This is direct evidence that the macro $Z_{L,t}$ and β_t time series do not pass through the household-level inversion. The aggregate wedges are economy-wide objects (utilization, expectations) that a household-level decomposition does not register, consistent with the discussion in Section 5.7.

Sorting peaks in prime working ages. Table 13 reports cross-parameter correlations within age bins, pooled across waves. In SCF, $\rho(\beta_i, z_i)$ is weak at age 25–29 (-0.18), strengthens to -0.34 at age 40–49, and weakens again at retirement age (-0.07 at 65–74). The wealth gradient $\rho(\beta_i, \log a_i)$ follows a similar inverted-U: -0.41 at 25–29, peaks at -0.54 at age 40–49, and weakens to -0.10 at age 65–74. PSID is flatter on $\rho(\beta_i, z_i)$ (close to zero across all bins) and on $\rho(\beta_i, \log a_i)$, consistent with the variance decomposition's finding that PSID's biennial saving identification has 70% idiosyncratic noise that dilutes cross-sectional correlations; the SCF intratemporal-FOC identification is sharper and reveals the inverted-U structure clearly. The sorting structure that drives the macro aggregate is strongest in prime working ages, when productivity differences manifest in wages, wealth differentials are largest, and life-cycle effects accumulate. Outside this window, sorting is weaker: at young ages most households are in early-career wealth-poor positions, while at retirement age wealth is decoupled from current labor productivity.

Table 13: Age-conditional cross-parameter correlations (waves pooled within survey)

Age bin	n	$\rho(\beta, z)$	$\rho(\beta, \log a)$	$\rho(z, \log a)$
PSID 2005–2019				
[25, 29]	2,199	+0.08	+0.01	+0.44
[30, 39]	5,804	+0.04	−0.04	+0.49
[40, 49]	5,954	−0.01	−0.08	+0.44
[50, 59]	6,159	−0.06	−0.08	+0.39
[60, 64]	2,029	−0.01	−0.09	+0.37
SCF 1992–2019				
[25, 29]	10,008	−0.18	−0.41	+0.18
[30, 39]	30,285	−0.27	−0.52	+0.27
[40, 49]	39,851	−0.34	−0.54	+0.30
[50, 59]	38,192	−0.30	−0.48	+0.25
[60, 64]	12,524	−0.32	−0.41	+0.19
[65, 74]	10,779	−0.07	−0.10	+0.05

Notes: Within-bin Pearson correlations, weighted by survey weight, pooled across waves. Both PSID and SCF β clipped at $[0.5, 1.5]$. Computed by `micro_diagnostics.py` from the panel CSVs.

5.6 Policy invariance: a Lucas-critique test

The tax-reform counterfactuals of Section 5.4 treat the structural parameters (z_i, e_i, β_i) as policy-invariant: the same households face the counterfactual tax sequence and respond through the closed-form MRCE policy at the new prices. The household-level inversion provides a direct test of that assumption around the four reforms whose enactment dates fall inside the SCF and PSID sample windows. This Lucas-test set (OBRA 1993, EGTRRA 2001, ATRA 2013, TCJA 2017) overlaps with but is not identical to the four reforms used in the macro counterfactual of Section 5.4 (Reagan ERTA 1981, OBRA 1993, EGTRRA 2001, ATRA 2013): the earliest macro reform (ERTA 1981) predates the micro panels, and the latest micro reform (TCJA 2017) had not yet propagated through the macro panel at the time of inversion. The two exercises therefore test the same policy-invariance hypothesis on a slightly shifted reform window.

Policy invariance: parameters stable around major tax reforms. The variance decomposition showed less than 0.3% of within-person z or β variance comoves with the wave. A direct policy test reinforces the same conclusion. The 27-year SCF window and 14-year PSID window contain four major federal tax reforms split between cuts and increases: the Omnibus Budget Reconciliation Act of 1993 (top rate 31% → 39.6%, an increase, between SCF 1992 and 1995); the Economic Growth and Tax Relief Reconciliation Act 2001 (a cut, between SCF 2001 and 2004); the American Taxpayer Relief Act 2013 (top rate 35% → 39.6% and capital-gains rate 15% → 20%, an increase, between SCF 2010–2013 and PSID 2011–2013); and the Tax Cuts and Jobs Act 2017 (corporate rate 35% → 21%, between SCF 2016–2019 and PSID 2017–2019). Table 14 reports the recovered moments in the wave just before and just after each reform.

Table 14: Parameter moments around major US tax reforms

Reform	Source	Pre–Post	β_{med}	β_{cap}	sd($\log z$)	$\rho(\beta, \log a)$
Tax increases						
OBRA 1993	SCF	1992–95	1.00 → 1.00	0.92 → 0.76	0.89 → 0.88	−0.55 → −0.50
ATRA 2013	SCF	2010–13	1.01 → 1.01	0.88 → 0.88	0.93 → 0.91	−0.42 → −0.42
ATRA 2013	PSID	2011–13	0.95 → 0.99	0.90 → 0.95	0.87 → 0.87	−0.07 → −0.01
Tax cuts						
EGTRRA 2001	SCF	2001–04	1.00 → 1.00	0.92 → 0.89	0.89 → 0.90	−0.55 → −0.48
TCJA 2017	SCF	2016–19	1.01 → 1.01	0.85 → 0.88	0.94 → 0.91	−0.47 → −0.46
TCJA 2017	PSID	2017–19	0.99 → 1.04	0.94 → 0.98	0.86 → 0.82	−0.12 → −0.02

Notes: Pre/post moments are computed within the consistent estimation sample for each survey, with population weights. The 1995 SCF β_{cap} outlier (0.76) reflects a wave-specific composition shift, not a sustained policy response: β_{cap} recovers to 0.91 by 1998. PSID changes are larger than SCF changes because the PSID biennial saving identification has wider β dispersion (Table 12: 70% of β variance is idiosyncratic noise). Computed by `micro_diagnostics.py` from the panel CSVs.

The cross-sectional distributions are essentially unchanged across all six pre/post comparisons. The median β moves by at most 0.06 in SCF; the standard deviation of $\log z$ is unaffected; the wealth gradient $\rho(\beta_i, \log a_i)$ stays in its narrow band. Differences are not systematically signed by the direction of the policy change.

The cleanest test uses PSID's within-person panel: compare the same household's β_i before and after each PSID-spanning reform. Across 2,955 households observed in both 2011 and 2013 (ATRA 2013 increase), mean $\Delta\beta_i = +0.017$ ($\sigma = 2.09$); across 3,162 households observed in both 2017 and 2019 (TCJA 2017 cut), mean $\Delta\beta_i = +0.021$ ($\sigma = 2.11$). The unconditional standard deviation of biennial $\Delta\beta_i$ across all 22,037 consecutive-wave pairs is 2.05, with mean −0.064. The policy-event averages lie well inside this typical noise band: individual households' recovered preferences do not shift around major tax reforms.

This is a Lucas-critique-style test that the structural identification passes. The macro inversion's tax counterfactuals (Section 5.4) treat structural parameters as policy-invariant, and the micro panel directly verifies that they are. Because the tested events span both increases and cuts, the policy-invariance result is not specific to the direction of any single political episode.

5.7 Macro and micro β as two scales

Macro–micro parallel and contrast. The two β's the paper recovers, namely the macro β_t from aggregate data and the micro β_i distribution from PSID and SCF, along with their $Z_{L,t}$ / z_i counterparts, play parallel but contrasting roles. The aggregate $Z_{L,t}$ is the time-varying common component of effective labor productivity (utilization, composition, matching), moving at business-cycle frequencies; individual z_i is the cross-sectional permanent characteristic. The aggregate β_t is the time-varying expectations / saving-margin wedge; individual β_i is the cross-sectional preference characteristic plus transitory wealth-shock noise. The macro inversion identifies the aggregate (time-series) component of each parameter; the micro inversion identifies the individual (cross-sectional) component. They are conceptually orthogonal: aggregate cyclical dynamics and cross-sectional heterogeneity. The framework's joint identification of both, via the same closed-form policies inverted at two scales, is the bridge between them.

Three threads pull the macro and micro inversions together.

Cross-survey replication is the strongest empirical content. PSID and SCF use orthogonal identification strategies on disjoint samples spanning three decades. PSID exploits a panel saving equation, identified from biennial wealth changes; SCF exploits a cross-sectional intratemporal labor FOC, identified from observed hours and wealth. The two recover capital-weighted $\hat{\beta}_{\text{cap}}$ within $\sim$1.7 pp of each other and signed-equal cross-sectional gradients in every wave. The MRCE structure (closed-form policies, time-invariant idiosyncratic processes, two-orthogonal-shocks decomposition) survives this independent replication. Two distinct datasets and two distinct identification routes pointing at the same answer is the strongest evidence the framework's micro side can produce.

Macro and micro β measure different economic objects. The macro β_t describes growth in universe productive capital (BEA fixed assets summed across all sectors); the micro β_i describes household balance-sheet dynamics. The systematic 7–9 pp gap between $\hat{\beta}_{\text{cap}} \approx 0.88$ and the macro $\beta_t = 0.967$ is dominated by the wealthy quartile: median Q4 wealth shrinks at -1.9%/yr in PSID, while universe-level capital grows at $+1\%$/yr in 2005–2019. The wedge is the household-vs-universe accumulation gap, not measurement error.[4]

[4] One might suspect housing wealth, with its asset-price volatility, accounts for the gap. The opposite holds: re-running the PSID inversion using PSID's `financial_wealth_real` alone (excluding housing equity)

Macro β as an expectations channel; cross-sectional β as preference heterogeneity. Section 4 reports a strong negative correlation $\rho(\Delta \log Z_{L,t}, \Delta\beta_t) = -0.858$ in macro first differences: aggregate patience rises sharply when labor productivity falls. The micro decomposition shows that this correlation does not arise at the household level. Cross-sectional $\rho(\beta_i, z_i)$ is small in any wave (below -0.05 in PSID, between -0.11 and -0.44 in SCF). The time series of micro $\hat{\beta}_{\text{cap,t}}$ co-moves only weakly with $Z_{L,t}$ at the wave frequency. Within-person regressions of $\Delta\beta_i$ on $\Delta Z_{L,t}$ are insignificant. Individual households do not literally become more patient when aggregate labor productivity falls. A natural reading is that the macro β_t captures aggregate *expectations* that co-move with $Z_{L,t}$: when labor productivity deteriorates, agents revise downward their expected paths of future income and returns, and aggregate saving rises in response, alongside corporate cash hoarding and other economy-wide forces that the household-level inversion does not register. The cross-sectional β_i heterogeneity, by contrast, identifies preference differences, and the within-person evidence above shows it has both a persistent (preference) component and a transitory (wealth-shock) component. The two objects, aggregate expectations and cross-sectional preferences, are conceptually distinct, and the micro inversion separates them.

Scope of the tax counterfactuals. The expectations reading of β above has a direct consequence for Section 5.4. The counterfactuals treat (z_i, e_i, β_i) as policy-invariant primitives, so any channel through which a new policy reshapes agents' expectations, and thereby their inferred patience, is by construction absent from the simulated paths. The Lucas-style test of Section 5.6 offers empirical comfort that the four reforms in our sample windows did not detectably move β_i, but the four episodes were moderate in scale and largely anticipated; the test does not speak to truly novel or unanticipated regimes for which an expectations response is plausible. The reported magnitudes are therefore best read as the sorting-channel response under the maintained policy-invariance hypothesis, with any expectations-induced movement in β_i left as a separate channel that would temper or amplify them depending on the sign of agents' belief revision.

6 Conclusion

This paper analyzes the distributional dynamics behind the observed aggregate dynamics via an analytical general equilibrium framework. In the model, aggregate capital is not the only

yields a wave-average $\hat{\beta}^{\text{fin}}_{\text{cap}} \approx 0.86$ and a wave-average gap of about -10 pp (range across waves: -22 to 0 pp), wider than the -7 pp gap from total net worth. The wedge is not driven by housing per se but by the broader mismatch between household balance-sheet dynamics and universe-level productive-capital growth; computed by `micro_diagnostics.py`.

relevant state; two sorting covariances also matter. They are sufficient for contemporaneous equilibrium allocations, they enter the aggregate capital law of motion in closed form, and they explain why representative-agent aggregation is accurate when sorting is quantitatively small.

A distinctive feature of the framework is that closed-form policies and belief invariance allow it to accept non-stationary data directly. Household decisions depend only on current prices, not on expectations about future aggregates. As a result, postwar U.S. output, capital, and hours are carried period by period by a single set of equilibrium equations, even as they trend through threefold growth while cycling through recurrent recessions and decades of rising wealth concentration. No detrending, filtering, or stationarity assumption is needed. Trend, cycle, and distributional change are jointly accounted for within one analytical structure rather than separated across different models or different filters.

The U.S. application carries three messages. First, the postwar business cycle is genuinely two-dimensional: TFP is nearly orthogonal to a labor-efficiency and patience block, so a single-shock representation loses identification rather than compressing it. Second, postwar wealth concentration is the endogenous accumulation of return-wealth sorting under TFP growth, not an exogenous distributional trend. The same three-moment structure that makes representative-agent aggregation accurate at baseline also generates decades of rising inequality from shocks alone. Third, the strength of the heterogeneous channel scales with shock persistence. Unconditional amplification and risk pricing are bounded by sorting strength, but persistent policy changes (capital-tax reforms in either direction) turn the same sorting mechanism into a decades-long distributional drift, with bidirectional symmetry across the four major postwar reforms (Reagan ERTA, OBRA 1993, Bush EGTRRA, ATRA 2013).

Extending the inversion to the household level closes the loop on the macro-side findings. The same closed-form policies recover individual structural parameters (z_i, e_i, β_i) from PSID and SCF micro data, taking the aggregate-inversion prices as given. Two results are first-order. First, two independent surveys with disjoint samples and orthogonal identification strategies agree on the capital-weighted aggregate $\hat{\beta}_{\text{cap}}$ within 1.7 pp, and both lie 7–9 pp below the macro β_t, a gap attributable to wealth-shock measurement at the individual level rather than to model failure. Second, neither household-level β_i nor z_i shifts around any of the four major federal tax reforms in the sample windows: a Lucas-critique-style test that the structural identification underlying the macro tax counterfactuals passes directly for the reforms tested (Section 5.7 discusses the scope of this result for novel regimes). Together, the macro and micro inversions establish that aggregate dynamics, distributional dynamics, and household decisions are governed by the same set of equations and the same set of recovered parameters at two different scales of aggregation.

Belief invariance is what makes these results practically accessible. The inversion at both scales, the rolling forecasts that beat a recursive AR benchmark for output and labor, and the tax counterfactuals are all obtained by forward simulation alone, without re-solving the household problem or taking a stand on expectations.

More broadly, the framework points toward inversion-based structural macroeconomics as a practical alternative to the conventional forward-solving paradigm. The dominant workflow in heterogeneous-agent macro has been to calibrate parameters, solve the household problem globally, simulate, and compare to data. The MRCE inverts this sequence: data in as the equilibrium path; prices, primitives, distributions, forecasts, and counterfactual paths out. When new data arrives, re-inversion is seconds rather than hours. Structural scenario analysis (e.g., "what if TFP drops 3 percent while the labor-efficiency and patience tandem activates?") requires only feeding a different shock sequence through the same forward simulation, with full general-equilibrium consistency and distributional consequences by construction.

Several avenues follow. First, embedding the MRCE in a New Keynesian setting would yield analytical characterizations of monetary policy interacting with distributional dynamics. Second, extending the inversion to higher-frequency (quarterly) data would bring the conditional forecasting framework closer to real-time policy use. Third, and most distinctive to the inversion approach, the same closed-form policies can be applied across countries: feeding each country's output, capital, and hours through the inversion recovers a comparable panel of structural objects: TFP, patience, and labor efficiency at the aggregate scale, and, where harmonized household surveys exist (for example the Luxembourg Wealth Study), the cross-sectional distribution of (z_i, e_i, β_i) at the micro scale. Because the recovered β_i is a structural valuation of the future that the inversion isolates from prices and returns, systematic cross-country differences in its level or its wealth gradient would be candidate measures of how institutions and culture shape patience, a quantity that forward-solving methods typically calibrate rather than recover. Cross-country inversion thus turns the framework into an instrument for detecting cultural and institutional heterogeneity in deep parameters, not merely in outcomes.

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